

AKTU



B. Tech I-Year

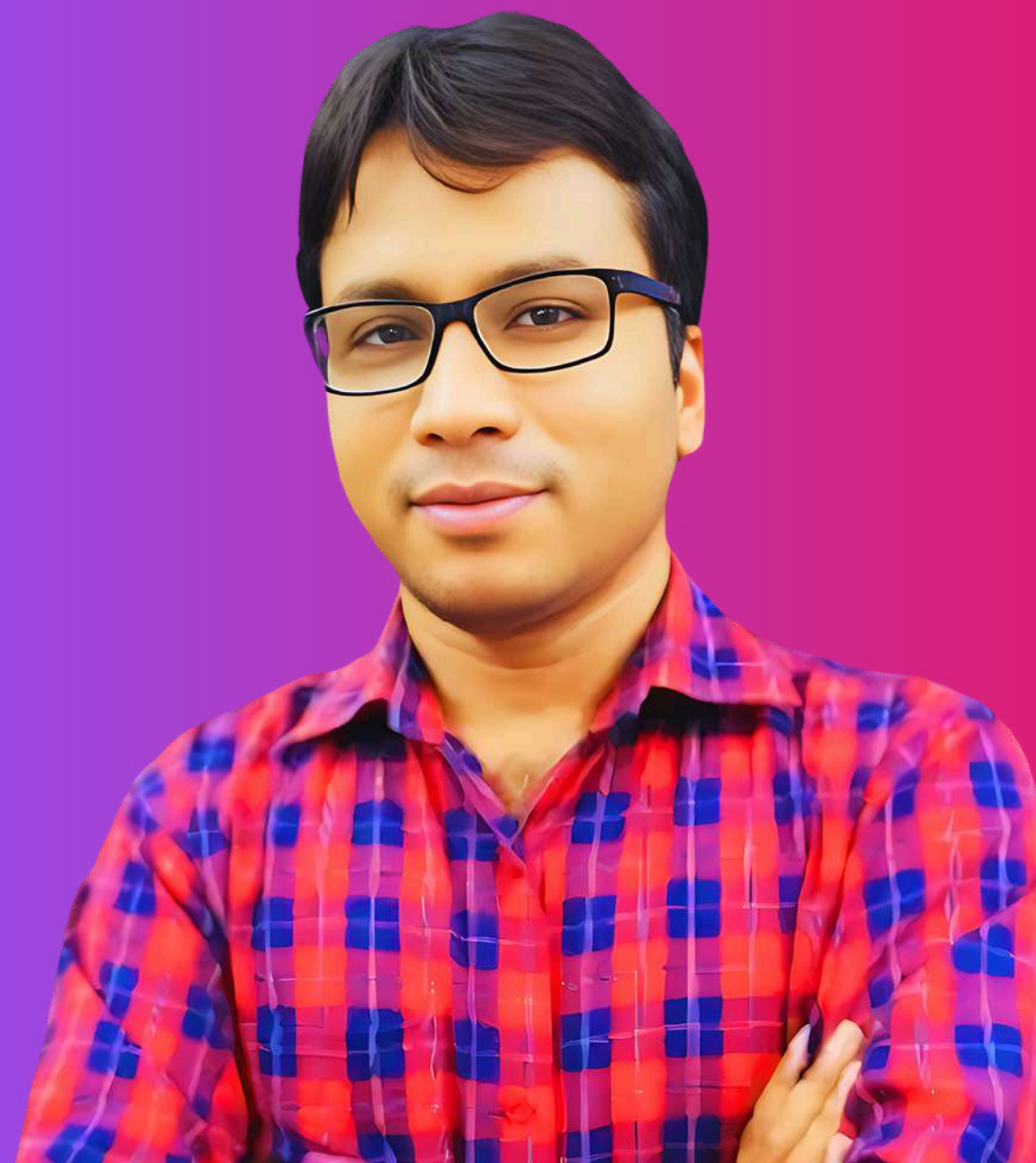
Gateway Classes



Semester -I & II **Common to All Branches**

BAS101 / BAS201: ENGINEERING PHYSICS

UNIT-3 (P-2) ONE SHOT : Diffraction



Gateway Series **for Engineering**

- ✓ **Topic Wise Entire Syllabus**
- ✓ **Long - Short Questions Covered**
- ✓ **AKTU PYQs Covered**
- ✓ **DPP**
- ✓ **Result Oriented Content**



Download App

For Full Courses including Video Lectures

AKTU



B. Tech I-Year

Gateway Classes



BAS101 / BAS201: ENGINEERING PHYSICS

Unit-3

Introduction to : Wave Optics (Diffraction) **Syllabus**

Coherent sources, Interference in uniform and wedge shaped thin films, Necessity of extended sources, Newton's Rings and its applications, Introduction to diffraction, Fraunhofer diffraction at single slit and double slit, Absent spectra, Diffraction grating, Spectra with grating, Dispersive power, Resolving power, Rayleigh's criterion of resolution, Resolving power of grating.



Download App

For Full Courses including Video Lectures





AKTU



Engg. Physics

ONE SHOT



Unit-3 : Wave Optics

Diffraction

By Gulshan Sir

One Shot PDF Notes Available on

Gateway Classes App

FREE



AKTU : B.Tech (II-Sem & IV-Sem)



S.No	COMBO PACK
1	B.Tech II-SEM COMBO
2	B.Tech IV-SEM : CS, IT & CS Allied
3	B.Tech IV-SEM : EC & EC Allied
4	B.Tech IV-SEM : ME & ME Allied
5	B.Tech IV-SEM : EE & EE Allied

Validity till semester exam

Helpline No. 7819 0058 53



AKTU : B.Tech (IV-Sem)



Engg. Maths-4 All Branches (Except CE/ENV)

Validity till semester exam

Helpline No. 7819 0058 53



AKTU : B.Tech



- Video Lectures
- Pdf Notes
- DPP
- PYQs
- Unit wise One Shot

Validity till semester exam

Helpline No. 7819 0058 53

Section – A

c. What happens if the slit is smaller than wavelength in diffraction pattern?

Section – B

c. Newton's rings are observed normally in reflected light of wavelength $6000\text{\AA}$. The diameter of the 10^{th} dark ring is 0.50cm . Find the radius of curvature of lens and thickness of the film.

Section – C

5. Attempt any one part of the following:

- (a) Discuss the phenomenon of interference of light due to parallel thin films and find the condition of maxima and minima. Show that the interference patterns of reflected and transmitted source of light are complementary.*
- (b) Discuss single slit Fraunhofer diffraction and show that the relative intensities of successive maximum are nearly $1: 1/22 : 1/62 : 1/121$.*

Section – A

d. What do you understand by coherent sources ?

Section – B

c. What do you understand by the phenomenon of Fraunhofer diffraction. Find out the ratio of intensities of successive secondary maxima compared to the intensity of the principle maximum.

Section – C

5. Attempt any one part of the following:

(a)(i) Describe the phenomenon of interference in thin film (uniform thickness) due to reflected light and write down the conditions for constructive and destructive interference.

(ii) A light source of wavelength $6000\text{\AA}$ is used along with plano – convex lens With radius of curvature equal to 100cm in a Newton's ring arrangement. Find out the diameter of the 15th dark ring.

(b) Explain briefly the Rayleigh criterion of resolution. Discuss the resolving power of plane transmission grating and find the relation between resolving and dispersive power of the grating.

Section – A

- d. Why two independent light sources cannot produce interference pattern?*
- e. What are the changes that are caused in the diffraction pattern if the number of slits are made large?*

Section – B

- c. Describe how Newton's ring experiment can be used to determine the refractive index of a liquid.*

Section – C

5. Attempt any one part of the following:

- (a) (i) Obtain an expression for the fringe width in a wedge – shaped thin film and explain nature of fringe pattern.*
- (ii) A light of wavelength $6000\text{\AA}$ falls normally on a slit of width 0.10 mm . Calculate the total angular width of the central maximum.*
- (b) (i) What particular spectra would be absent if width of the transparencies and opacities of the grating are equal.*
- (ii) A plane transmission grating has 16,000 lines to an inch over a length of 5 inches. Find in wavelength region of $6000\text{\AA}$, in the second order, the smallest wavelength difference that can be resolved.*

Section – A

- g.** The light rays from two independent bulbs do not show interference. Give the reason.
- h.** State the Rayleigh criteria of resolution.

Section – B

- d.** White light is incident on a soap film at an angle $\sin^{-1} (4/5)$ and the reflected light is observed with a spectroscope. It is found that two consecutive dark bands correspond to wavelengths $6.1 \times 10^{-5} \text{ cm}$ and $6.0 \times 10^{-5} \text{ cm}$. If the refractive index of the film is $4/3$, calculate the thickness.

Section – C

6. Attempt any one part of the following:

- (a)** What do you mean by a wedge-shaped film? Discuss the interference due to it and obtain the expression for the fringe width.
- (b)** Discuss the formation of Newton's rings. Show that the diameters of the bright rings are proportional to the square root of odd natural numbers.

Section – A

g. Why the center of Newton's ring in reflected system is dark?

h. Explain Rayleigh's criterion of resolution.

Section – B

d. Newton's rings are observed in reflected light of wavelength $5900\text{\AA}$. The diameter of 10th dark ring is 0.50cm . Find the radius of curvature of the lens.

6. Attempt any one part of the following:

(a) What is Rayleigh criterion of resolution how one can increase the resolving power of a diffraction grating? Using Rayleigh criterion for just resolution show that the resolving power of grating is equal to nN , where n is the order of the spectrum, and N is total no of lines on the grating.

(b) Discuss the phenomena of Fraunhofer diffraction at a single slit and show that the relative intensities of the successive maximum are nearly $1 : 4/9\pi^2 : 4 / (25\pi^2) : 4/49\pi^2 : \dots$

Section – A

- g.** Two independent sources of light cannot produce interference, why?
- h.** State Rayleigh criterion of Resolution. Also define resolving power.

Section – B

- d.** Calculate the thickness of a soap bubble thin film that will result in constructive interference in reflected light. The film is illuminated with light of wavelength $5000 \text{ \AA}$ and the refractive index of the film is 1.45.

Section – C

6. Attempt any one part of the following:

- (a)** Describe the formation of Newton's rings in monochromatic light. Show that in reflected light, the diameters of dark rings are proportional to the square roots of natural numbers.
- (b)** What is a diffraction grating? Discuss the phenomenon of diffraction due to plane diffraction grating.

Section – A

- f.** Two independent sources could not produce interference, why ?
- g.** What is dispersive power of plane transmission grating ?

Section – B

- d.** A plane transmission grating has 16,000 lines to an inch over a length of 5 inches. Find in the wavelength region of $6000 \text{ \AA}$, in the second order (i) the resolving power of grating and (ii) the small wavelength difference that can be resolved.

Section – C

6. Attempt any one part of the following:

- (a)** Describe and explain the formation of Newton's rings in reflected monochromatic light. Deduce the necessary expression for bright and dark rings.
- (b)** Discuss the phenomenon of Fraunhofer diffraction at a single slit. Show that the intensity of the first subsidiary maximum is about 4.5% of the principal maximum.



B. Tech : Engg. Physics



Unit : Wave Optics



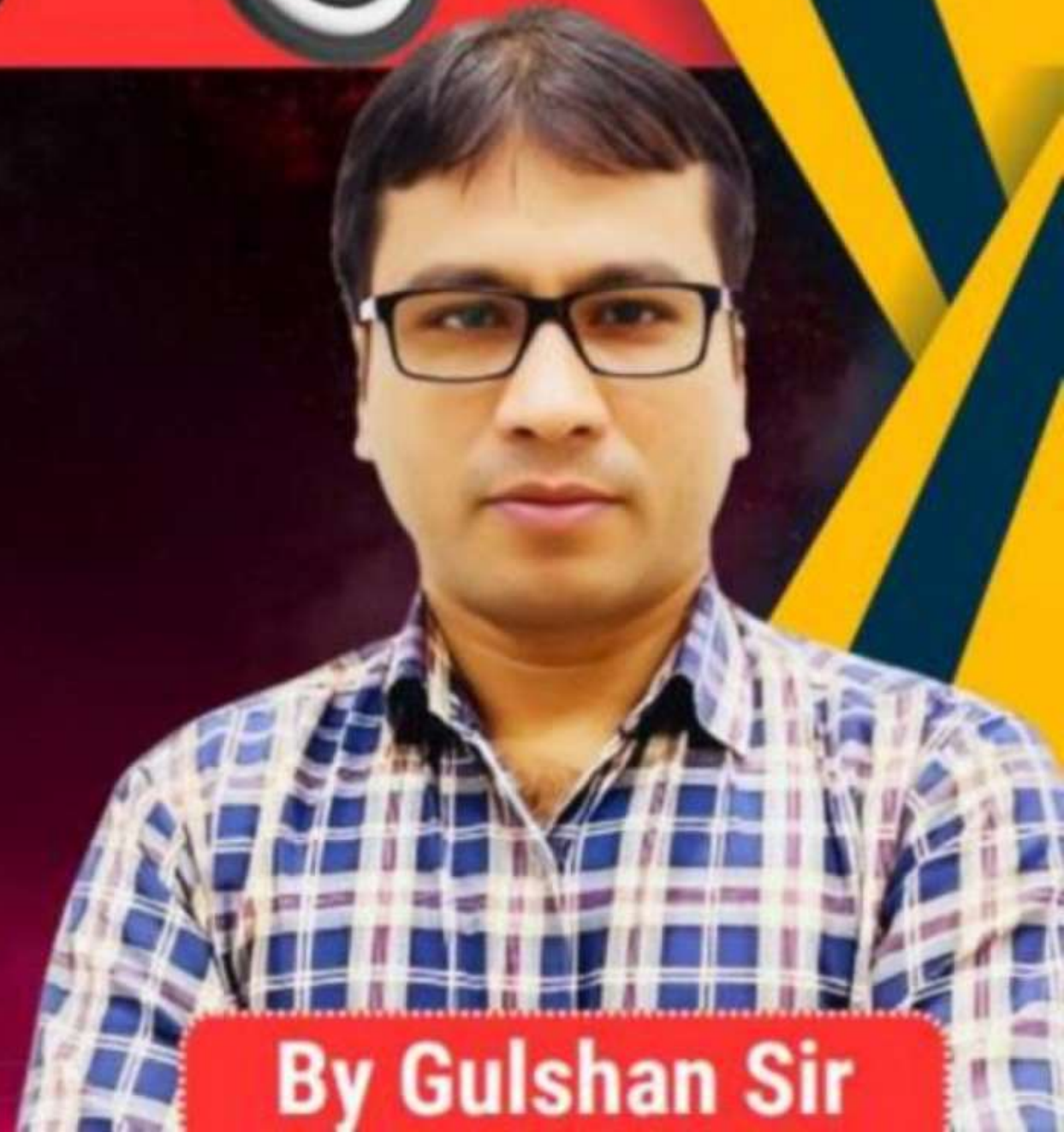
(Diffraction)

Lecture - 5



Today's Target

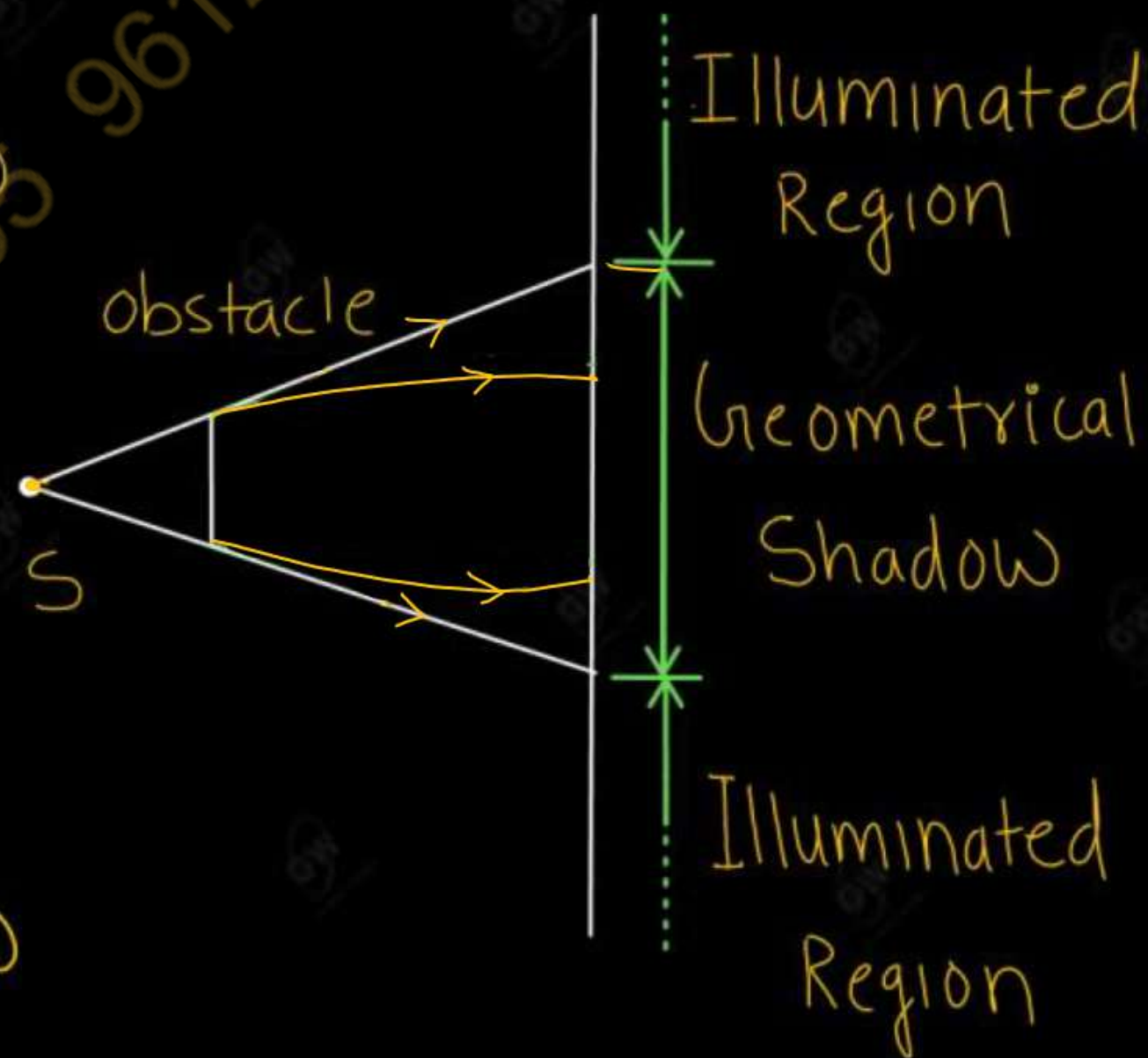
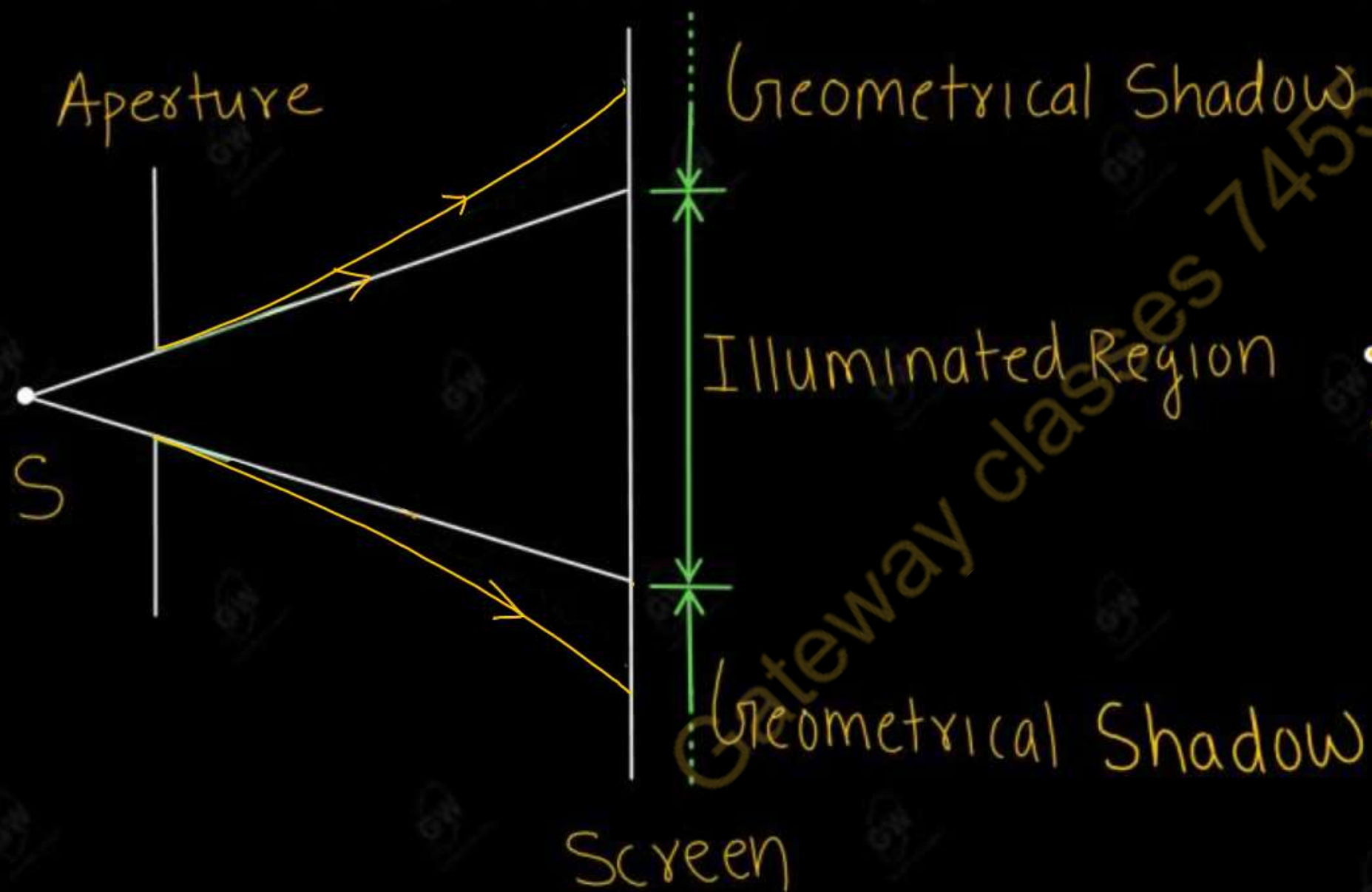
- ✓ Introduction to Diffraction
- ✓ Fraunhofer diffraction at single slit V. imp
- ✓ DPP
- ✓ PYQs



By Gulshan Sir

- When light falls on an opaque obstacle or aperture whose size is comparable with the wavelength of light, the light bends around the corners of the obstacle or aperture and enter in to the geometrical shadow.

This bending of light is called Diffraction.



- If the size of the obstacle compared to the wavelength of the light wave is
 - (i) Very small (i.e $a \ll \lambda$), then the wave will undergo reflection and not diffraction
 - (ii) Very large (i.e $a \gg \lambda$), then the wave will not be diffracted
 - (iii) Almost equal (i.e $a \approx \lambda$), then the wave will undergo maximum diffraction
- Hence, the phenomenon of Diffraction depends on the size of the obstacle (a) and the wavelength of the light (λ).
- There are two types of diffraction
 - (i) Fraunhofer diffraction ✓
 - (ii) Fresnel diffraction

Gateway classes 7455 9612 87

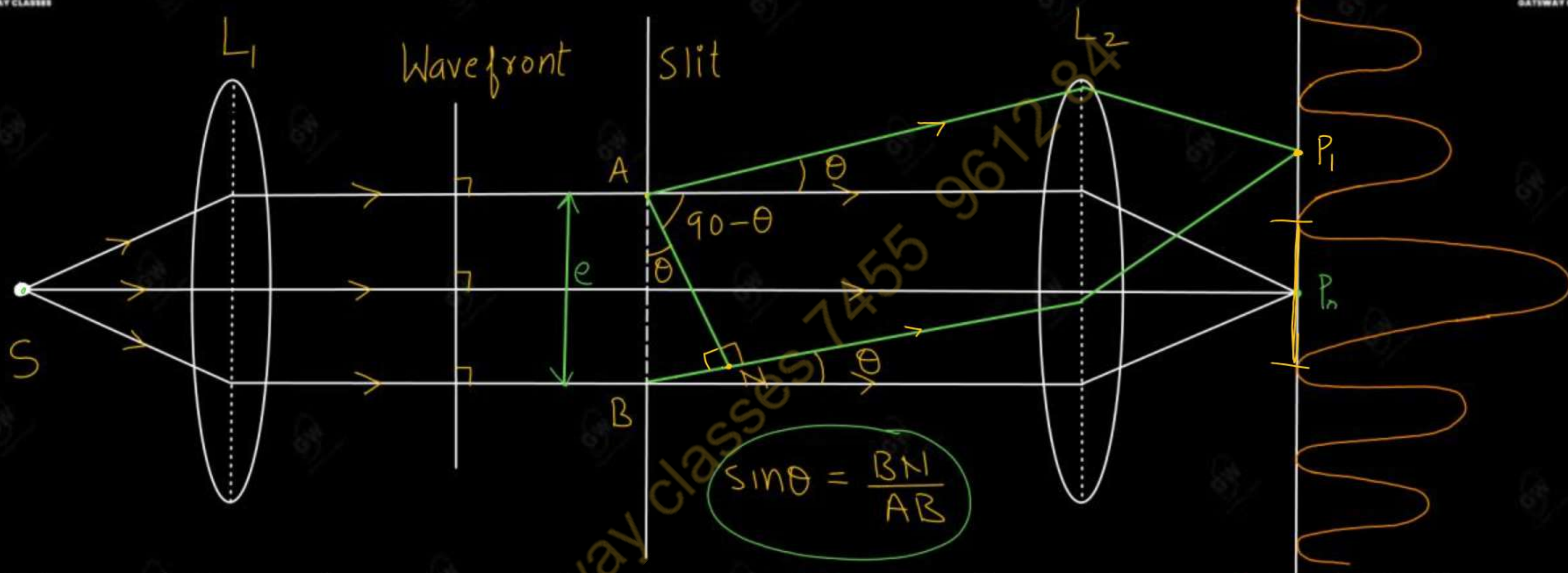
Difference between Fraunhofer diffraction and Fresnel diffraction

Fraunhofer diffraction

- In this class of diffraction, source and screen are placed at infinite distance from the diffracting aperture
- The wave front is plane
- The Centre of diffraction pattern is always bright
- In this class of diffraction, two convex lens are used to observe the diffraction pattern

Fresnel diffraction

- In this class of diffraction, source and screen are placed at finite distance from the diffracting aperture
- The wave front is either spherical or cylindrical
- The Centre of diffraction pattern is either bright or dark
- In this class of diffraction, lenses or mirror are not used



- Let a parallel beam of monochromatic light of wavelength λ produced by a point source S placed at the focus of a convex lens (L_1) be incident normally on a narrow slit AB of width e where it gets diffracted.
- If a converging lens L_2 is placed in the path of diffracted beam, a real image of the diffraction pattern is formed on the screen in the focal plane of the lens.

Path Difference (Δ)

$$\Delta = BN$$

$$\Delta = AB \sin \theta$$

$$\Delta = e \sin \theta$$

Phase Difference (δ)

$$\text{Total Phase Difference} = \frac{2\pi}{\lambda} \Delta$$

$$\text{Total phase difference} = \frac{2\pi e \sin \theta}{\lambda}$$

If slit is divided into 'n' parts then according to Huygen's wave theory each part become an elementary source of light and amplitude of wave from each part is 'a'

Now, phase difference between any two consecutive waves

$$\delta = \frac{2\pi e \sin \theta}{n \lambda} \quad \text{--- (1)}$$

From the theory of n Harmonic
Vibration

The Resultant amplitude is given by

$$R = \frac{a \sin\left(\frac{n\delta}{2}\right)}{\sin\left(\frac{\delta}{2}\right)}$$

Using ①

$$R = \frac{a \sin\left(\frac{\cancel{n} \times \cancel{2} \times \pi e \sin \theta}{\cancel{2} \times \cancel{n} \lambda}\right)}{\sin\left(\frac{\cancel{2} \times \pi e \sin \theta}{\cancel{2} \times n \lambda}\right)}$$

$$R = \frac{a \sin\left(\frac{\pi e \sin \theta}{\lambda}\right)}{\sin\left(\frac{\pi e \sin \theta}{n \lambda}\right)}$$

Put $\frac{\pi e \sin \theta}{\lambda} = \alpha$ ——— ②

$$R = \frac{a \sin \alpha}{\sin(\alpha/n)}$$

$\therefore \frac{\alpha}{n}$ is very very small

$$\therefore \sin \frac{\alpha}{n} = \frac{\alpha}{n}$$

$$R = \frac{a \sin \alpha}{\frac{\alpha}{n}}$$

$$R = \frac{an \sin \alpha}{\alpha}$$

Put $an = A$

$$R = \frac{A \sin \alpha}{\alpha}$$

Resultant Intensity

$$I = (\text{Amplitude})^2$$

$$I = R^2$$

$$I = \frac{A^2 \sin^2 \alpha}{\alpha^2}$$

$$I = A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

③

Position and condition for
Central Maxima

Intensity will be maximum, When

$$\frac{\sin \alpha}{\alpha} = 1$$

$$\text{But } \lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} = 1$$

$$\text{Now } I_0 = A^2 \left(\lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} \right)^2$$

$$I_0 = A^2 (1)^2$$

$$I_0 = A^2$$

When

$$\alpha = 0$$

$$\frac{\pi e \sin \theta}{\lambda} = 0 \quad \{\text{using (2)}\}$$

$$\Rightarrow \sin \theta = 0$$

$$\theta = 0^\circ$$

Thus, intensity will be maximum, when $\theta = 0$.

This maximum intensity is called central maxima or principal maxima

Position and condition for minima

Intensity will be minimum, when

$$\sin \alpha = 0 \text{ and } \alpha \neq 0$$

$$\alpha = \pm n\pi \quad n = 1, 2, 3, \dots$$

$$\frac{\pi e \sin \theta}{\lambda} = \pm n\pi$$

$$e \sin \theta = \pm n\lambda \quad \text{V. imp}$$

Position and condition for secondary Maxima

$$\frac{dI}{d\alpha} = 0$$

$$I = A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

diff. wrt α

$$\frac{dI}{d\alpha} = A^2 \times 2 \left(\frac{\sin \alpha}{\alpha} \right) \times \left(\frac{\alpha \cos \alpha - \sin \alpha}{\alpha^2} \right)$$

$$0 = 2A^2 \left(\frac{\sin \alpha}{\alpha} \right) \left(\frac{\alpha \cos \alpha - \sin \alpha}{\alpha^2} \right)$$

$$\Rightarrow \frac{\alpha \cos \alpha - \sin \alpha}{\alpha^2} = 0$$

$$\alpha \cos \alpha - \sin \alpha = 0$$

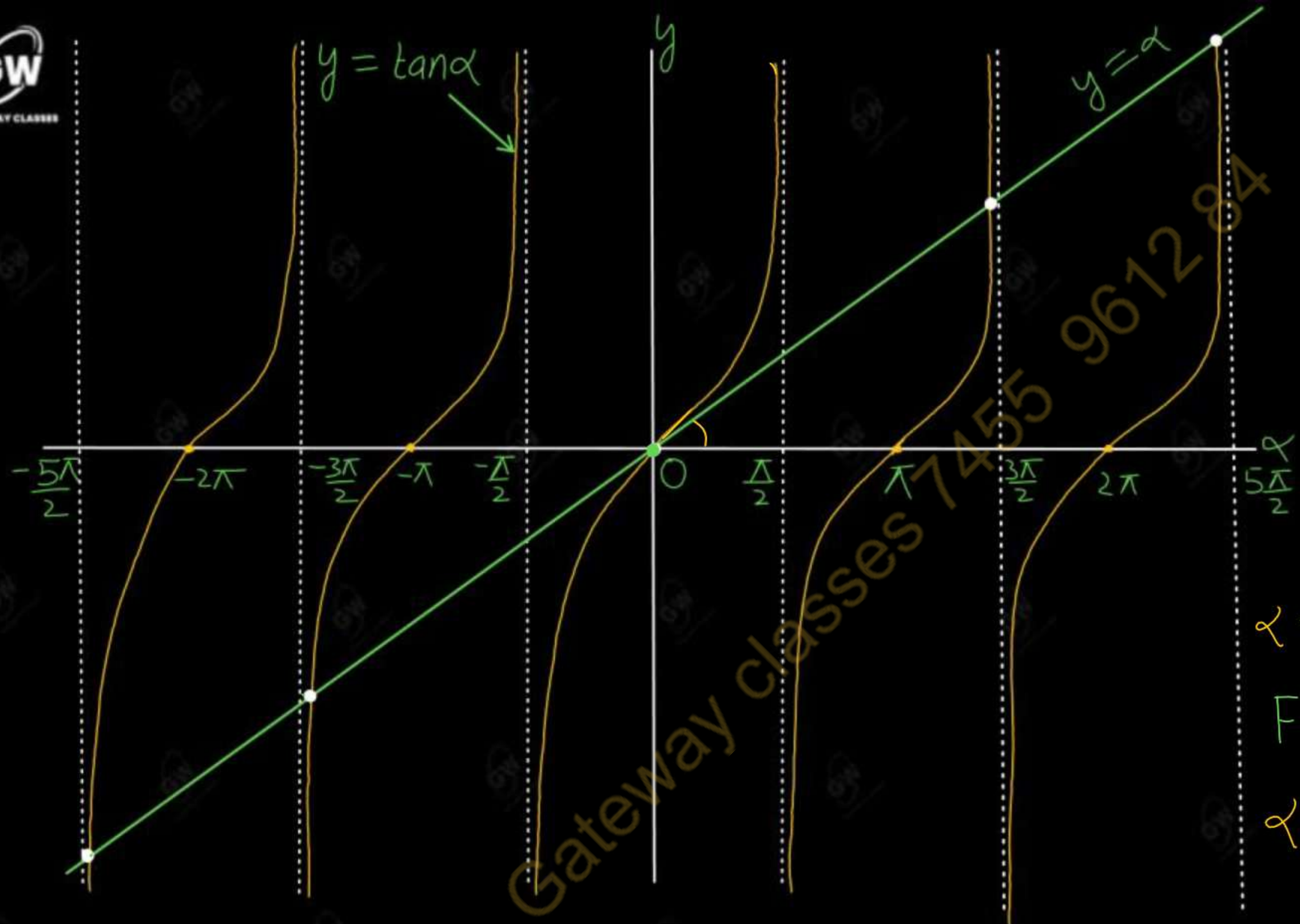
$$\alpha \cos \alpha = \sin \alpha$$

$$\alpha = \frac{\sin \alpha}{\cos \alpha}$$

$$\alpha = \tan \alpha$$

Let $y = \alpha$ and $y = \tan \alpha$

Now, plot the graph of $y = \alpha$
and $y = \tan \alpha$



Points of intersection

$$\alpha = 0, \pm \frac{3\pi}{2},$$

$$\pm \frac{5\pi}{2}, \pm \frac{7\pi}{2} \dots$$

But

$\alpha = 0$ (central Maxima)

For secondary Maxima

$$\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}$$

$$\pm \frac{9\pi}{2} \dots$$

$$\alpha = \pm (2n+1) \frac{\pi}{2} \quad n=1, 2, 3, \dots$$

$$\frac{\cancel{\lambda}}{\lambda} e \sin \theta = \pm (2n+1) \frac{\cancel{\lambda}}{2}$$

v.imp
$$e \sin \theta = \pm (2n+1) \frac{\lambda}{2} \quad n=1, 2, 3, \dots$$

Intensity of First Secondary Maxima

$$I_1 = \frac{A^2 \sin^2 \alpha}{\alpha^2}$$

$$I_1 = I_0 \frac{\sin^2 3\pi/2}{\left(\frac{3\pi}{2}\right)^2}$$

$$I_1 = \frac{4}{9\pi^2} I_0$$

$$I_1 = \frac{1}{22} I_0$$

Thus intensity

of first secondary maxima is 4.5% of principal maxima

Intensity of second secondary Maxima

$$I_2 = \frac{A^2 \sin^2 \alpha}{\alpha^2} = \frac{I_0 \sin^2 5\pi/2}{\left(\frac{5\pi}{2}\right)^2}$$

$$I_2 = \frac{4}{25\pi^2} I_0$$

$$I_2 = \frac{1}{62} I_0$$

Thus, intensity of Secondary Maxima is about 1.61% of Central Maxima

Third Secondary Maxima

$$I_3 = \frac{A^2 \sin^2 \alpha}{\alpha^2}$$

$$I_3 = \frac{I_0^2 \sin^2 7\pi/2}{\left(\frac{7\pi}{2}\right)^2}$$

$$I_3 = \frac{4}{49\pi^2} I_0$$

$$I_3 = \frac{1}{121} I_0$$

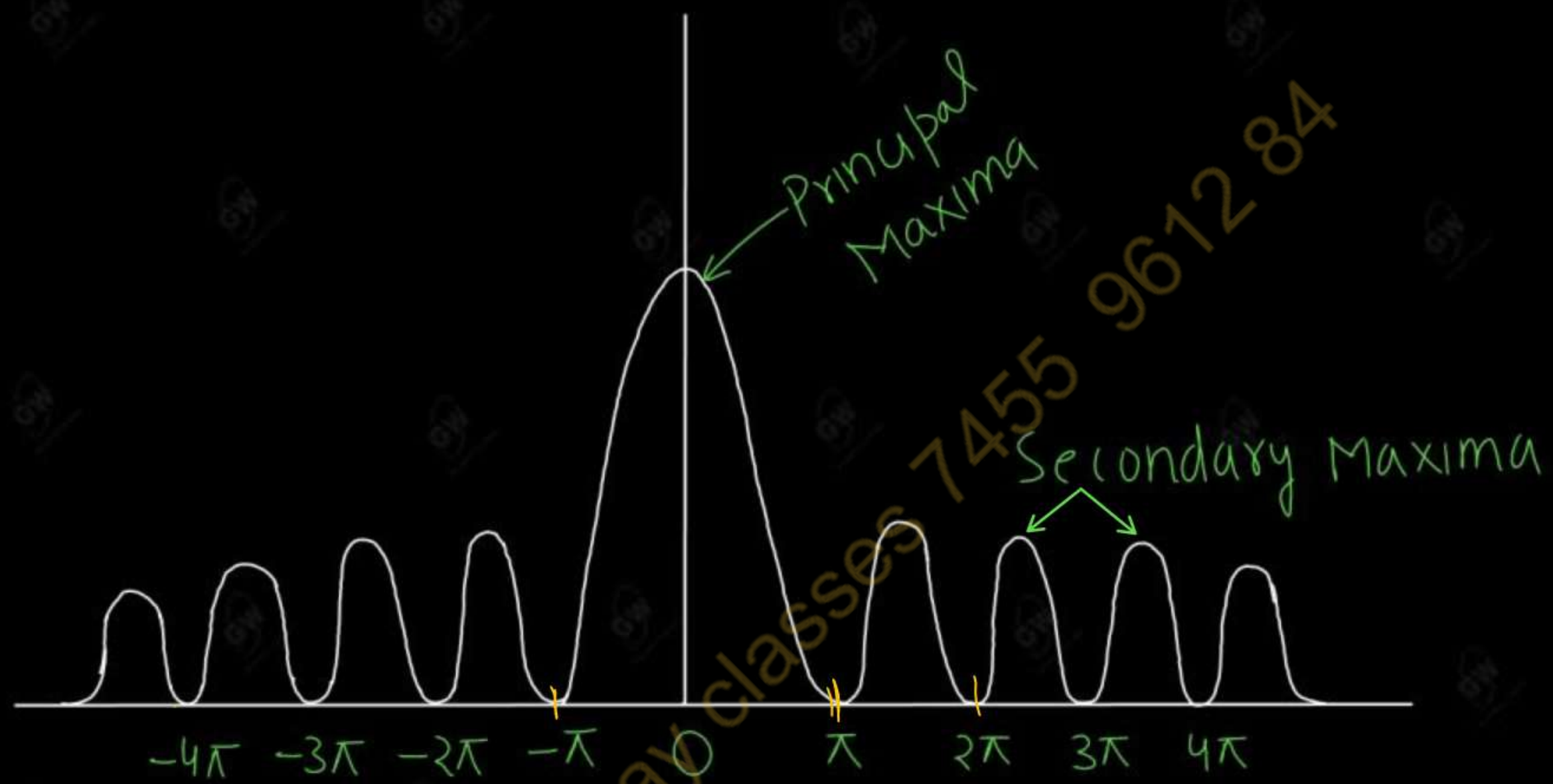
Thus secondary maxima decreases rapidly

Ratio of intensities of Secondary maxima

$$= 1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2} : \dots$$

OR

$$= 1 : \frac{1}{22} : \frac{1}{62} : \frac{1}{121} : \dots$$



Intensity Distribution due to Single Slit

Difference between interference and diffraction

Interference

- Interference is due to interaction of light from two different wave fronts originating from same source.
- Interference fringes are of the same width
- All bright fringes are of the same intensity
- All points of minimum intensity are perfectly dark
- The spacing between fringes is uniform

Diffraction

- Diffraction is due to interaction of light coming from different parts of the same wave front.
- Diffraction fringes are not of same width.
- All bright fringes are not of the same intensity.
- All points of minimum intensity are not perfectly dark
- The spacing between fringes is not uniform.

Q.1 Light of wavelength 5500 Å falls normally on a slit of width 22.0×10^{-5} cm. Calculate the angular position of the first two minima on either side of the central maximum.

Given

$$\lambda = 5500 \text{ Å}$$

$$e = 22 \times 10^{-5} \text{ cm}$$

Condition for minima

$$e \sin \theta = n \lambda$$

$$\sin \theta = \frac{n \lambda}{e}$$

(i) First Minima ($n=1$)

$$\sin \theta = \frac{1 \times 5500 \times 10^{-8}}{22 \times 10^{-5}}$$

$$\sin \theta = 0.25$$

$$\theta = 14.47^\circ$$

(ii) Second Minima ($n=2$)

$$\sin \theta = \frac{2 \times 5500 \times 10^{-8}}{22 \times 10^{-5}}$$

$$\sin \theta = 0.5$$

$$\theta = 30^\circ$$

Q.2 A light of wavelength $6000 \text{ \AA}$ falls normally on a strength slit of width 0.10 mm .

Calculate the total angular width of the central maximum and also the linear width as observed on a screen placed 1 metre away.

Given

$$\lambda = 6000 \text{ \AA} = 6000 \times 10^{-8} \text{ cm}$$

$$e = 0.10 \text{ mm} = 0.01 \text{ cm}$$

$$n = 1$$

$$D = 100 \text{ cm}$$

condition for minima

$$e \sin \theta = n \lambda$$

$$\sin \theta = \frac{n \lambda}{e}$$

For First minima ($n=1$)

$$\sin \theta = \frac{1 \times 6000 \times 10^{-8}}{0.01 \text{ cm}}$$

$$\sin \theta = 6 \times 10^{-3}$$

$\therefore \theta$ is very-very small

$$\sin \theta \approx \theta$$

$$\theta = 6 \times 10^{-3} \text{ radian}$$

(i) Total Angular width of central Maxima is 2θ

$$2\theta = 2 \times 6 \times 10^{-3}$$

$$2\theta = 1.2 \times 10^{-2} \text{ radian}$$

(ii) Linear Width of central Maxima is $D \times 2\theta$

$$D \times 2\theta = 100 \times 1.2 \times 10^{-2}$$

$$D \times 2\theta = 1.2 \text{ cm}$$

Q.3 Calculate the angle at which the first dark band and the next bright band are formed
in the Fraunhofer diffraction pattern of a slit 0.3 mm wide ($\lambda = 5890\text{\AA}$)

Given

$$\lambda = 5890 \text{\AA}$$

$$e = 0.3 \text{ mm} = 0.03 \text{ cm}$$

Direction of minima

$$e \sin \theta = n \lambda$$

$$\sin \theta = \frac{n \lambda}{e}$$

(i) For First Dark band
 $n = 1$

$$\sin \theta = \frac{1 \times 5890 \times 10^{-8}}{0.03}$$

$$\theta = 0.112^\circ$$

(ii) First Bright band on
either side of central
maxima

OR

First Secondary maxima ($n=1$)

$$e \sin \theta = (2n+1) \frac{\lambda}{2}$$

$$\sin \theta = \frac{3 \lambda}{2e}$$

$$\sin \theta = \frac{3 \times 5890 \times 10^{-8}}{2 \times 0.03}$$

$$\theta = 0.168^\circ$$

DPP-5

UNIT : Wave Optics

Q.1 What do you understand by the phenomenon of Fraunhofer diffraction. Find out the ratio of intensities of successive secondary maxima compared to the intensity of the principle maximum.

OR

Discuss the phenomena of Fraunhofer diffraction at a single slit and show that the relative intensities of the successive maximum are nearly $1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} : \frac{4}{49\pi^2} : \dots$

OR

Discuss single slit Fraunhofer diffraction and show that the relative intensities of successive maximum are nearly $1 : \frac{1}{22} : \frac{1}{61} : \frac{1}{121} \dots$

OR

Discuss the phenomenon of Fraunhofer diffraction at a single slit. Show that the intensity of the first subsidiary maximum is about 4.5% of the principal maximum.



B. Tech : Engg. Physics



Unit : Wave Optics

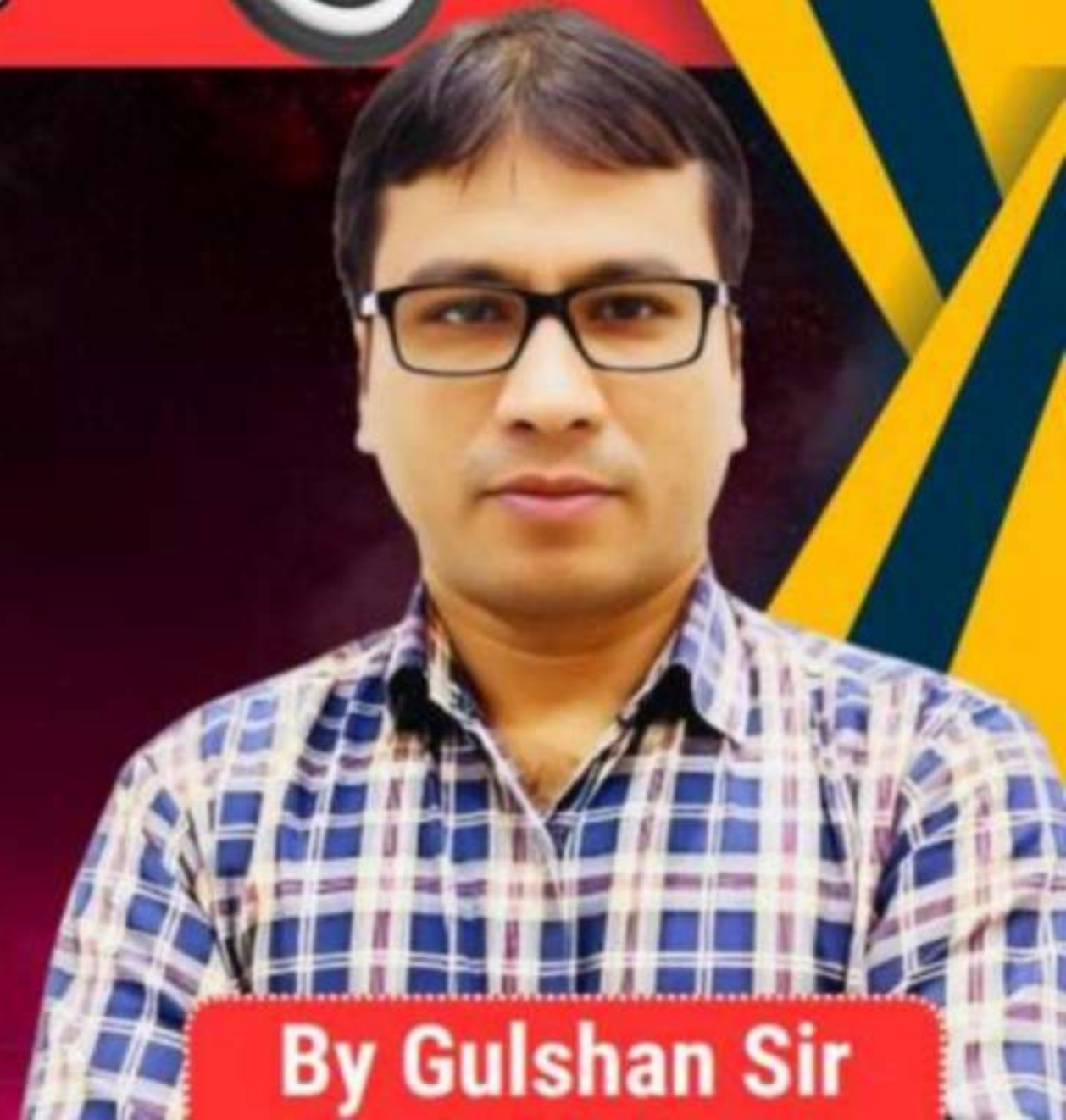


(Diffraction)

Lecture - 6

Today's Target

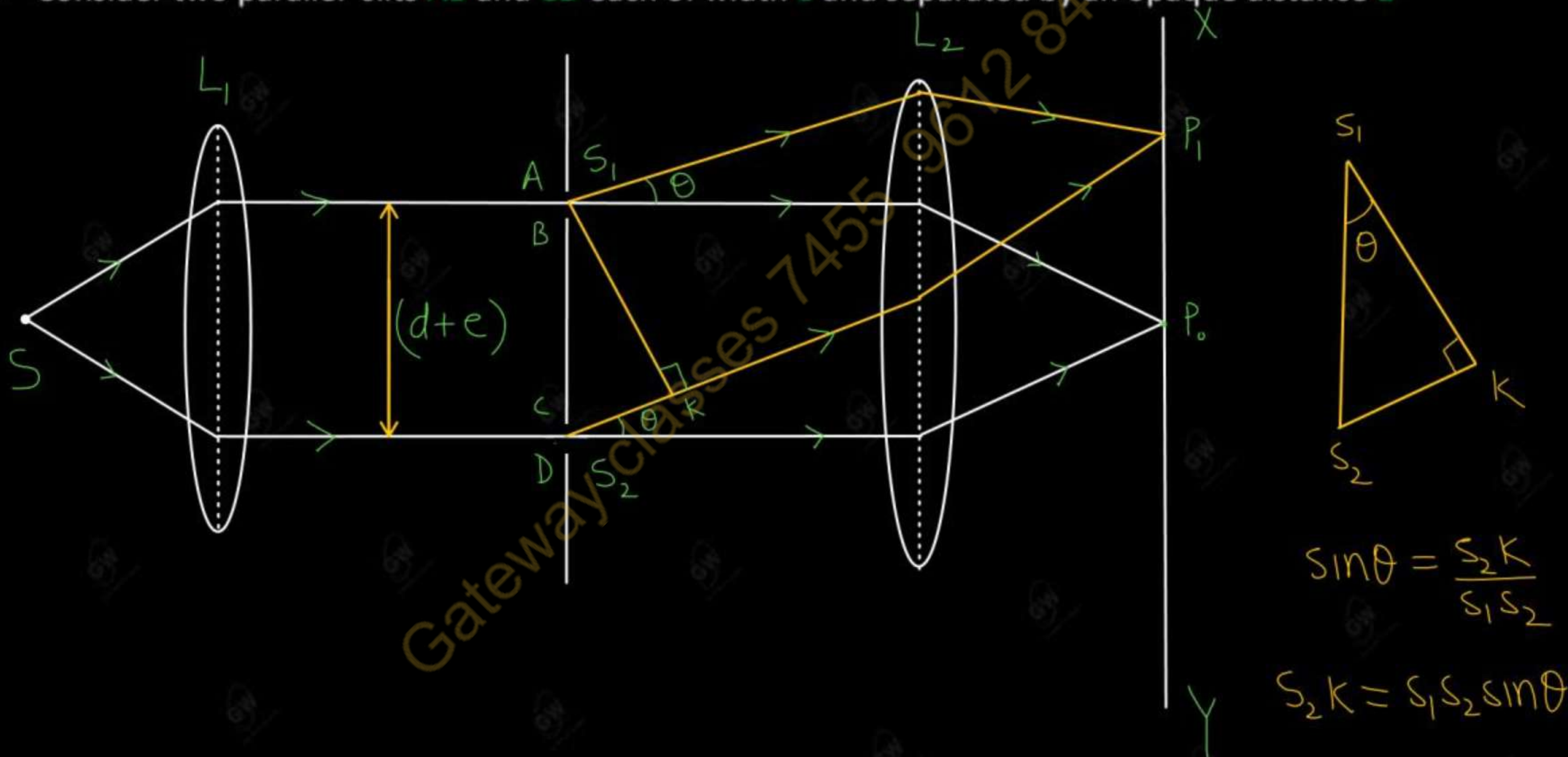
- ✓ Fraunhofer diffraction at Double Slit
- ✓ DPP
- ✓ PYQs



By Gulshan Sir

Fraunhofer diffraction at Double Slit

- Consider two parallel slits AB and CD each of width e and separated by an opaque distance d



Path Difference (Δ)

$$\Delta = s_2 \lambda$$

$$\Delta = s_1 s_2 \sin \theta$$

$$\Delta = (d+e) \sin \theta$$

Phase difference (δ)

$$\delta = \frac{2\pi}{\lambda} \Delta$$

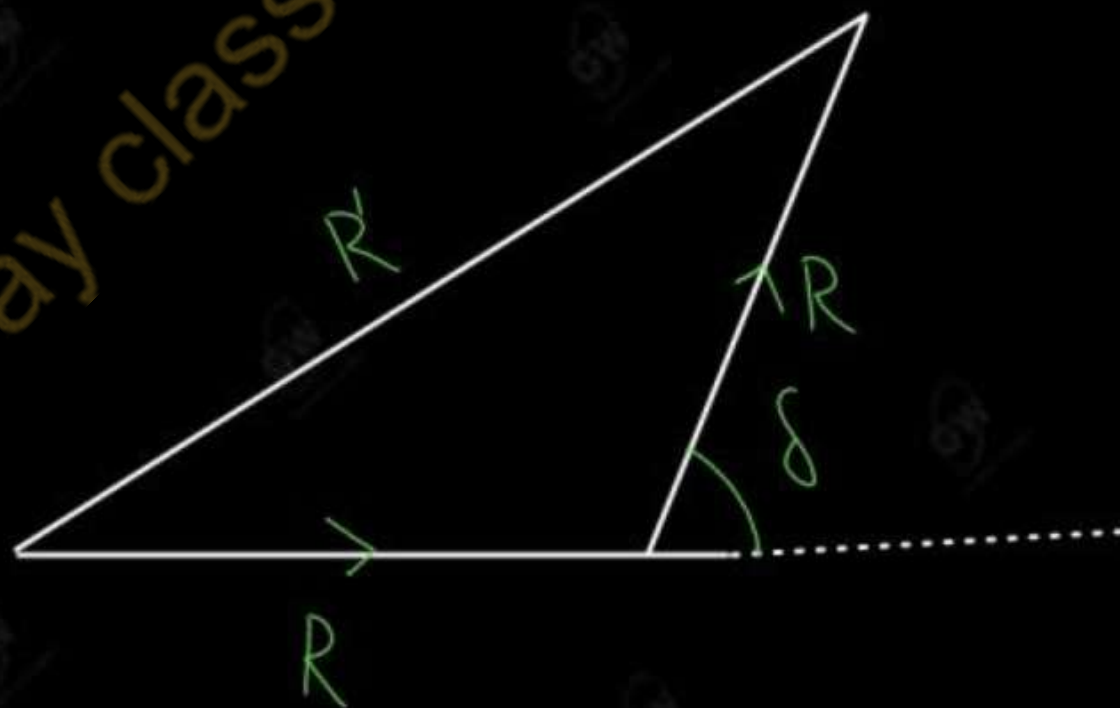
$$\delta = \frac{2\pi}{\lambda} (d+e) \sin \theta$$

Resultant Amplitude for slit s_1 and s_2

$$R = \frac{A \sin \alpha}{\alpha} \quad \left\{ \because \text{By single slit diffraction} \right\}$$

Where $\alpha = \frac{\pi e \sin \theta}{\lambda}$

Resultant Amplitude at P_1



Where

$$R = \frac{A \sin \alpha}{\alpha}$$

By parallelogram law of vector
Addition

$$R'^2 = R^2 + R^2 + 2R \times R \cos \delta$$

$$R'^2 = 2R^2 + 2R^2 \cos \delta$$

$$R'^2 = 2R^2 (1 + \cos \delta)$$

$$R'^2 = 2R^2 2 \cos^2 \delta/2$$

$$R'^2 = 4R^2 \cos^2 \left\{ \frac{2\pi (d+e) \sin \theta}{\lambda} \right\}$$

$$R'^2 = 4R^2 \cos^2 \left\{ \frac{\pi (d+e) \sin \theta}{\lambda} \right\}$$

Let $\frac{\pi (d+e) \sin \theta}{\lambda} = \beta$

$$R'^2 = 4R^2 \cos^2 \beta$$

$$R'^2 = 4 \left(\frac{A \sin \alpha}{\alpha} \right)^2 \cos^2 \beta$$

$$R'^2 = 4A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$$

Resultant Intensity

$$I = R'^2$$

$$I = 4A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta$$

↓
Gives Diffraction
Pattern

↓
Gives Interference
pattern

Diffraction Pattern

(i) Condition for central Maxima

$$\alpha = 0$$

$$\frac{\pi}{\lambda} e \sin \theta = 0$$

$$\sin \theta = 0$$

$$\theta = 0$$

$$I_{\max} = 4A^2$$

(ii) condition for minima

$$\sin \alpha = 0 \text{ and } \alpha \neq 0$$

$$\sin \alpha = \sin n\pi \text{ and } \alpha = 0$$

$$\alpha = \pm n\pi \quad n=1, 2, 3$$

$$\frac{\lambda}{\lambda} e \sin \theta = n\lambda$$

$$e \sin \theta = n\lambda$$

$$I_{\min} = 0$$

(iii) condition for Secondary maxima

$$\alpha = \tan \alpha$$

Where

$$\alpha = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \pm \frac{7\pi}{2}, \dots$$

$$\alpha = \pm (2n+1)\frac{\pi}{2}$$

$$\frac{\lambda}{\lambda} e \sin \theta = \pm (2n+1)\frac{\lambda}{2}$$

$$e \sin \theta = \pm (2n+1)\frac{\lambda}{2}$$

Interference Pattern

(i) condition for maxima

$$\cos^2 \beta = 1$$

$$\beta = \pm m\pi \quad m = 0, 1, 2$$

$$\frac{\cancel{\lambda}}{\lambda} (e+d) \sin \theta = \pm m \cancel{\lambda}$$

$$(e+d) \sin \theta = \pm m\lambda$$

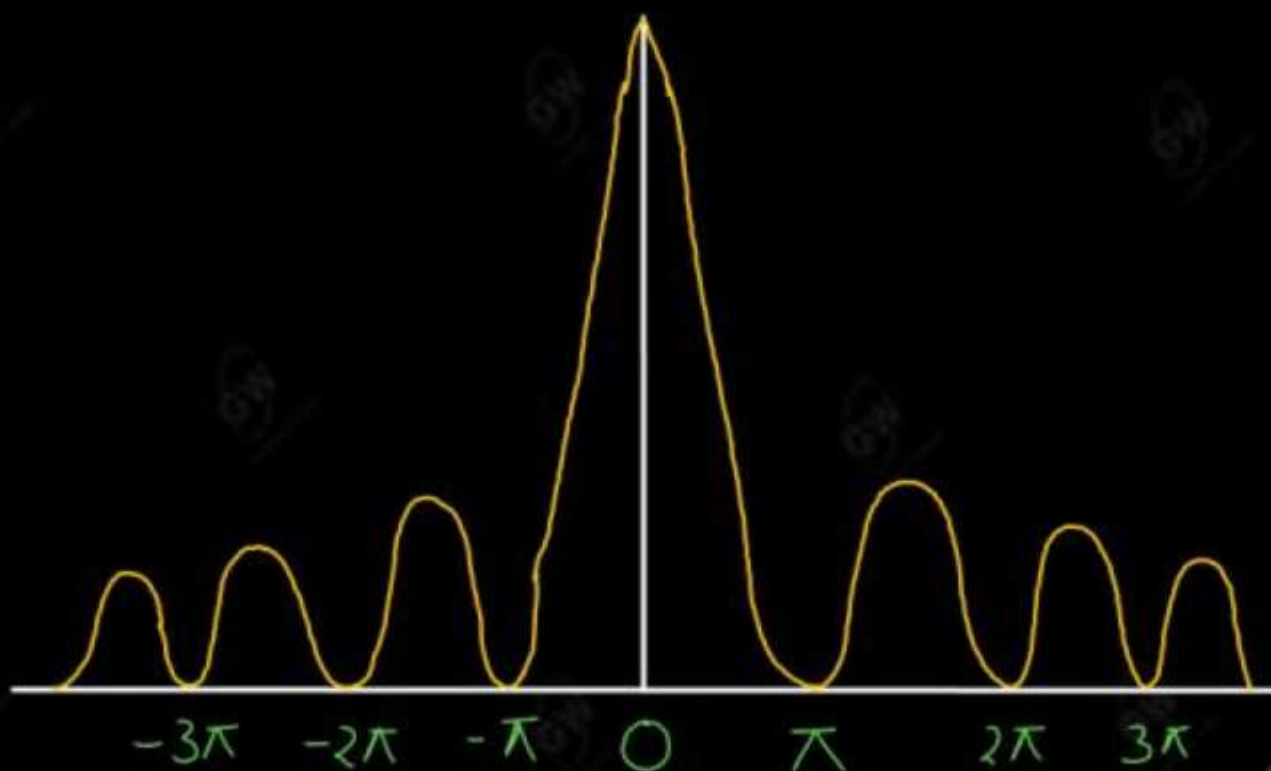
(ii) condition for minima

$$\cos^2 \beta = 0$$

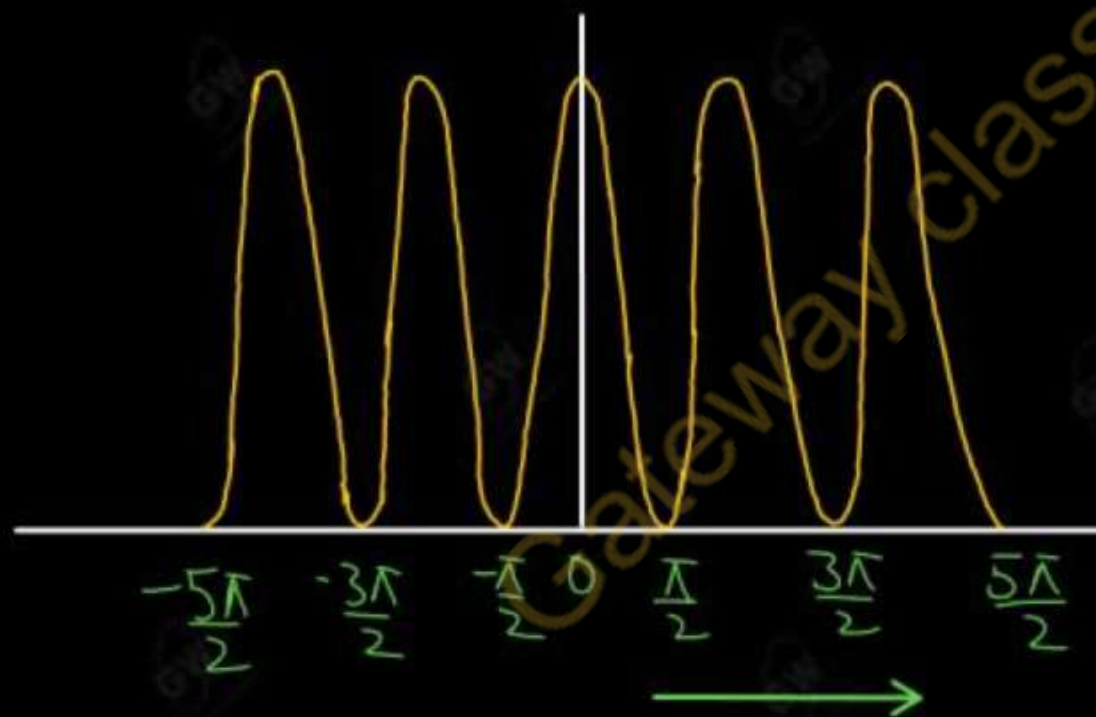
$$\beta = \pm (2m+1) \frac{\pi}{2}$$

$$\frac{\cancel{\lambda}}{\lambda} (e+d) \sin \theta = \pm (2m+1) \frac{\cancel{\lambda}}{2}$$

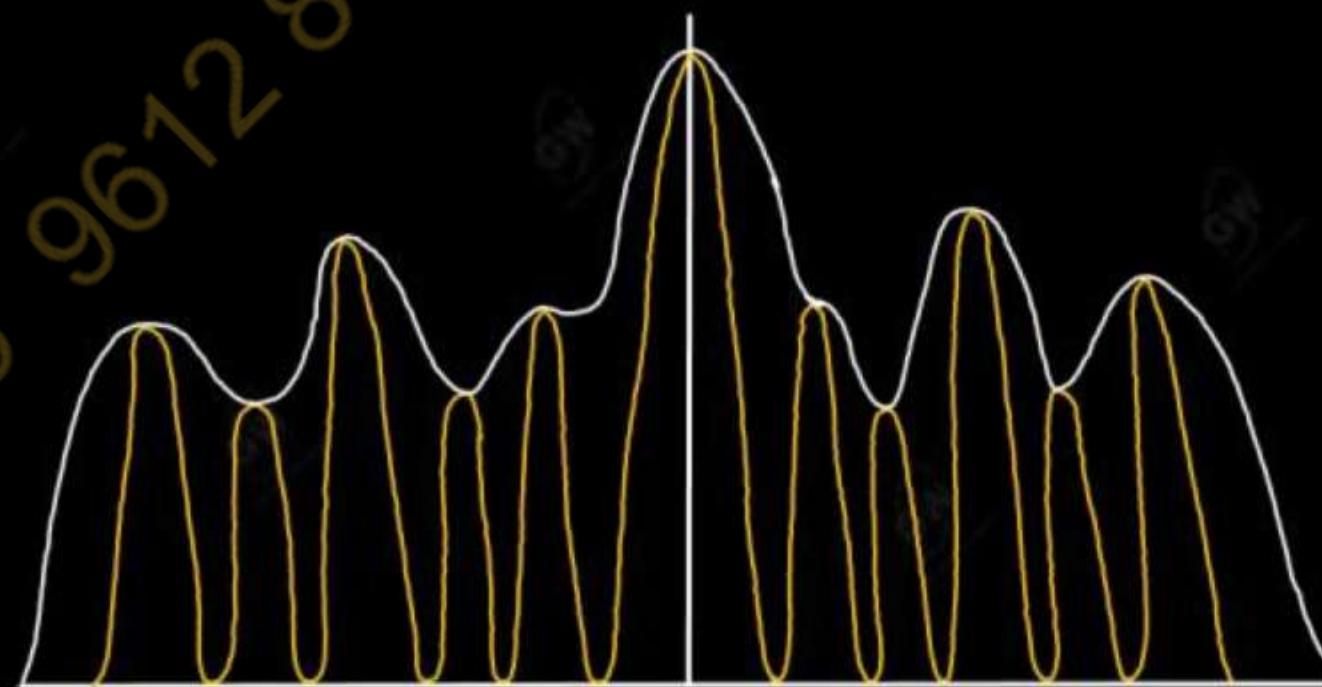
$$(e+d) \sin \theta = \pm (2m+1) \frac{\lambda}{2}$$



$\alpha \longrightarrow$



$\longrightarrow$



$\theta \longrightarrow$

- The direction of interference maxima are given by

$$(e + d)\sin\theta = m\lambda \quad \text{--- (1)}$$

- The direction of diffraction minima are given by

$$e \sin\theta = n\lambda \quad \text{--- (2)}$$

Divide (1) by (2)

$$\frac{(e+d)\cancel{\sin\theta}}{e\cancel{\sin\theta}} = \frac{m\cancel{\lambda}}{n\cancel{\lambda}}$$

$$\boxed{\frac{e+d}{e} = \frac{m}{n}}$$

- (i) If $e = d$

$$\frac{2\cancel{d}}{\cancel{d}} = \frac{m}{n}$$

$$m = 2n$$

If $n = 1, 2, 3, \dots$

$$m = 2, 4, 6, \dots \dots \dots \text{(Missing order)}$$

This means that 2, 4, 6, Orders of the interference maxima will be missing in the diffraction pattern.

- (ii) If $2e = d$

$$\frac{3\cancel{e}}{\cancel{e}} = \frac{m}{n}$$

$$m = 3n$$

$$n = 1, 2, 3, \dots$$

$$m = 3, 6, 9, \dots \dots \dots \text{(Missing Order)}$$

This means that 3, 6, 9, Orders of the interference maxima will be missing in the diffraction pattern.

Q.1 Discuss the phenomena of Fraunhofer diffraction by Double Slits

Q.2 What will be the effect on the intensity of principle maxima of diffraction pattern when single slit is replaced by double slit.

Gateway classes 7455 96284



B. Tech : Engg. Physics



Unit : Wave Optics



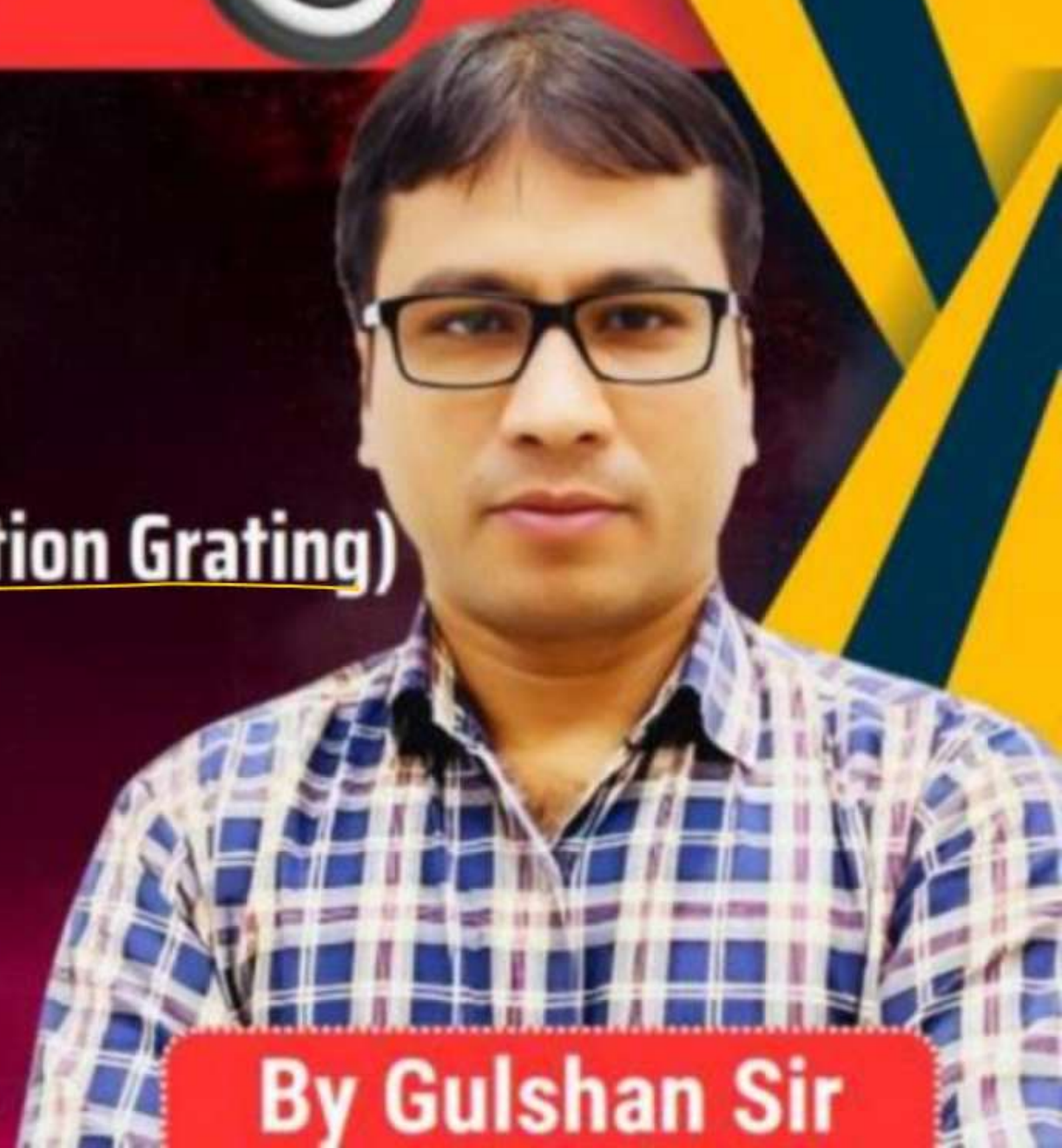
(Diffraction)

Lecture - 7



Today's Target

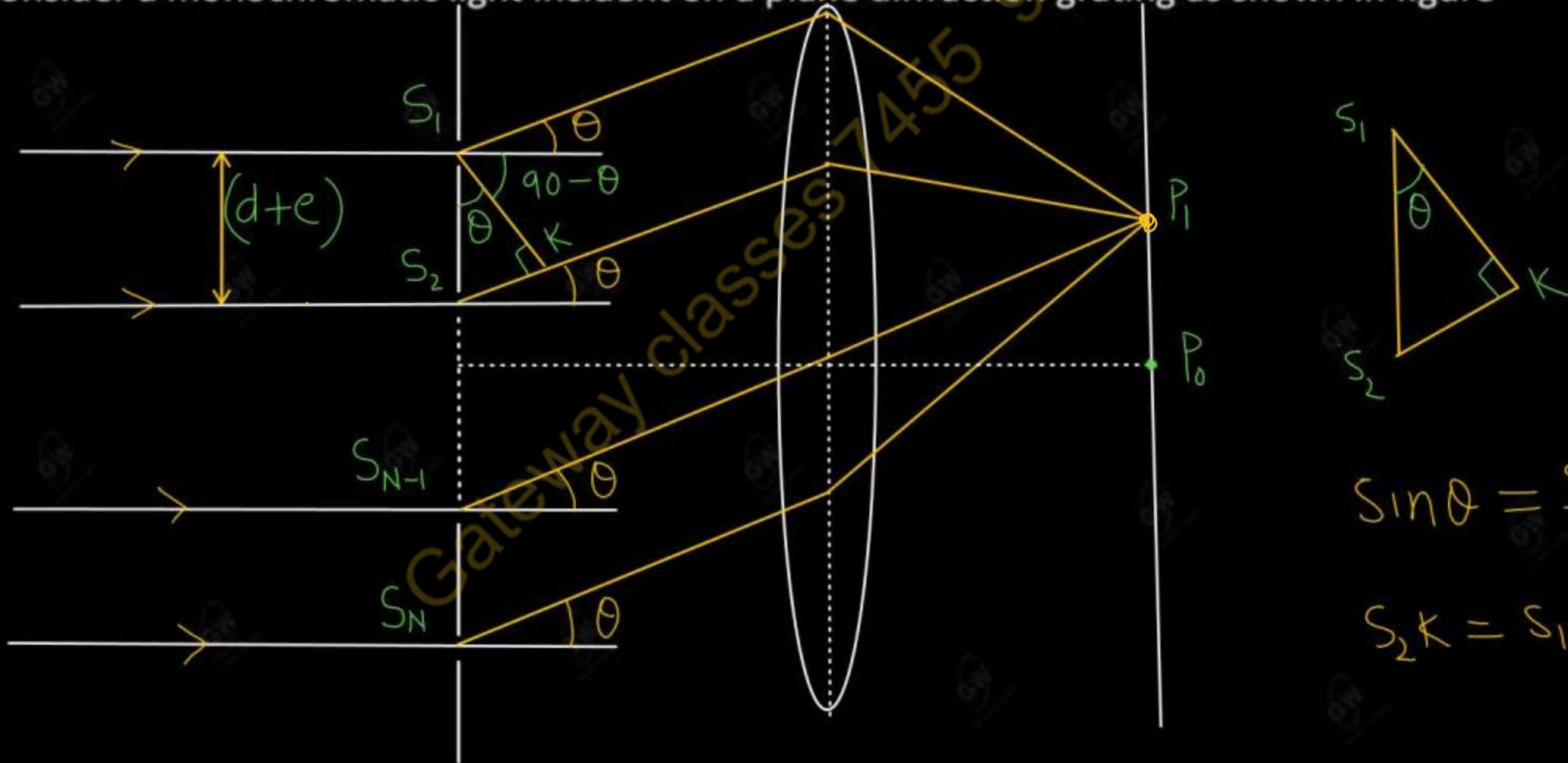
- ✓ Fraunhofer diffraction by N Parallel Slits(or Diffraction Grating)
- ✓ Absent Spectra , Spectra with grating
- ✓ DPP
- ✓ PYQs



By Gulshan Sir

Fraunhofer diffraction Due to N Parallel Slits (Diffraction Grating)

- An arrangement consisting of a large number of parallel slits (N) of same width (e) and separated by the same opaque space (d) is known as the diffracting grating
- The distance between the two consecutive slits ($e + d$) is known as grating element
- Consider a monochromatic light incident on a plane diffraction grating as shown in figure



$$\sin \theta = \frac{S_2K}{S_1S_2}$$

$$S_2K = S_1S_2 \sin \theta$$

Path Difference (Δ)

$$\Delta = s_1 s_2 \sin \theta$$

$$\Delta = (e+d) \sin \theta$$

Phase Difference (δ)

$$\delta = \frac{2\pi}{\lambda} \Delta$$

$$\delta = \frac{2\pi}{\lambda} (e+d) \sin \theta$$

$$\beta = \frac{2\pi}{\lambda} (e+d) \sin \theta$$

$$\beta = \frac{\pi}{\lambda} (e+d) \sin \theta$$

Resultant Amplitude for each slit

$$R = \frac{A \sin \alpha}{\alpha}$$

Where

$$\alpha = \frac{\pi}{\lambda} e \sin \theta$$

Resultant Amplitude at P_1
due to N Waves (From N Slits)

$$R' = \frac{R \sin\left(\frac{N\delta}{2}\right)}{\sin\left(\frac{\delta}{2}\right)}$$

But $\delta = 2\beta$ and $R = \frac{A \sin \alpha}{\alpha}$

$$R' = \frac{A \sin \alpha}{\alpha} \times \frac{\sin N\beta}{\sin \beta}$$

Resultant Intensity

$$I = R'^2$$

$$I = \left(\frac{A^2 \sin^2 \alpha}{\alpha^2} \right) \times \left(\frac{\sin N\beta}{\sin \beta} \right)^2 \quad \text{--- (1)}$$

Intensity due to
diffraction at single
Slit

Intensity due to
Interference of light
Waves from all
 N slits

Position and condition for central Maxima

Intensity will be maximum, when

$\frac{\sin N\beta}{\sin \beta}$ is maximum

i.e. $\sin \beta = 0$

$$\beta = \pm n\pi$$

, $n = 0, 1, 2, 3, \dots$

$$\lim_{\beta \rightarrow \pm n\pi} \frac{\sin N\beta}{\sin \beta}$$

Using L-Hospital Rule

$$\lim_{\beta \rightarrow \pm n\pi} \frac{N \cos N\beta}{\cos \beta} = N$$

$$I_{\max} = \frac{A^2 \sin^2 \alpha}{\alpha} \times (N)^2$$

$$I_{\max} = A^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \times N^2 \quad \text{--- (2)}$$

$$I_{\max} \propto N^2$$

Direction of central maxima

$$\beta = \pm n\pi$$

$$\cancel{\frac{\pi}{\lambda}} (e+d) \sin \theta = \pm n \cancel{\pi}$$

V.V.I.M.P

$$(e+d) \sin \theta = \pm n \lambda \quad n = 0, 1, 2, 3, \dots$$

→ Bragg's Equation

Note

- (i) $n = 0$ (Zero order principal Maxima)
 - (ii) $n = 1$ (First " " " " ")
 - (iii) $n = 2$ (Second " " " " ")
- and soon

Condition for minima

Intensity will be minimum, when

$$\sin N\beta = 0 \text{ and } \sin \beta \neq 0$$

$$N\beta = \pm m\pi$$

$$\cancel{N} \frac{\pi}{\lambda} (e+d) \sin \theta = \pm m \cancel{\pi}$$

$$N(e+d) \sin \theta = \pm m \lambda \quad \checkmark$$

$$m = 1, 2, 3, 4, \dots, N-1$$

$$m \neq 0, N, 2N, 3N, \dots$$

Hence, there are $(N-1)$ minima between two maxima

Condition for Secondary maxima

$$\frac{dI}{d\beta} = 0$$

$$I = \left(\frac{A^2 \sin^2 \alpha}{\alpha^2} \right) \times \left(\frac{\sin N\beta}{\sin \beta} \right)^2$$

diff. w.r.t β

$$\frac{dI}{d\beta} = \frac{A^2 \sin^2 \alpha}{\alpha^2} \frac{d}{d\beta} \left(\frac{\sin N\beta}{\sin \beta} \right)^2$$

$$0 = \left(\frac{A^2 \sin^2 \alpha}{\alpha^2} \right) \times 2 \left(\frac{\sin N\beta}{\sin \beta} \right) \times \left[\frac{N \sin \beta \cos N\beta - \sin N\beta \cos \beta}{\sin^2 \beta} \right]$$

$$\left(\frac{A \sin \alpha}{\alpha} \right)^2 \times 2 \left(\frac{\sin N\beta}{\sin \beta} \right) \left[\frac{N \sin \beta \cos N\beta - \sin N\beta \cos \beta}{\sin^2 \beta} \right] = 0$$

$$\frac{N \sin \beta \cos N\beta - \sin N\beta \cos \beta}{\sin^2 \beta} = 0$$

$$N \sin \beta \cos N\beta - \sin N\beta \cos \beta = 0$$

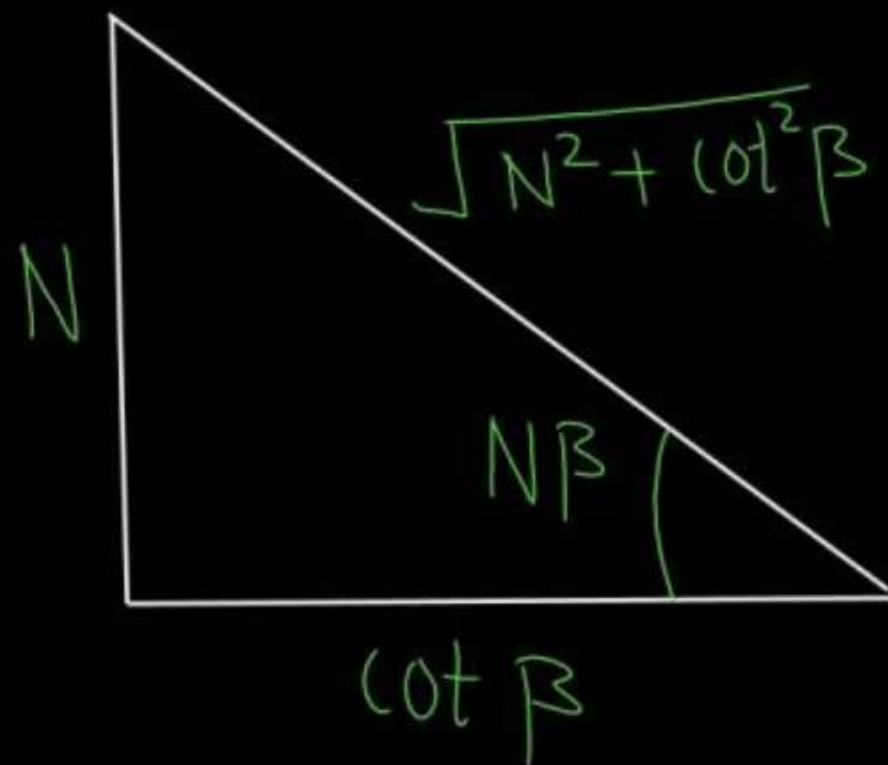
$$N \sin \beta \cos N\beta = \sin N\beta \cos \beta = 0$$

$$N \frac{\sin \beta}{\cos \beta} = \frac{\sin N\beta}{\cos N\beta}$$

$$N \tan \beta = \tan N\beta$$

$$\tan N\beta = N \tan \beta$$

$$\tan N\beta = \frac{N}{\cot \beta}$$



$$\sin N\beta = \frac{N}{\sqrt{N^2 + \cot^2 \beta}}$$

$$\sin^2 N\beta = \frac{N^2}{N^2 + \cot^2 \beta}$$

Divide by $\sin^2 \beta$

$$\frac{\sin^2 N\beta}{\sin^2 \beta} = \frac{N^2}{\sin^2 \beta (N^2 + \cot^2 \beta)}$$

$$\left(\frac{\sin N\beta}{\sin \beta} \right)^2 = \frac{N^2}{N^2 \sin^2 \beta + \cancel{\sin^2 \beta} \times \frac{\cos^2 \beta}{\cancel{\sin^2 \beta}}}$$

$$\left(\frac{\sin N\beta}{\sin \beta} \right)^2 = \frac{N^2}{N^2 \sin^2 \beta + \cos^2 \beta}$$

$$= \frac{N^2}{N^2 \sin^2 \beta + 1 - \sin^2 \beta}$$

$$\left(\frac{\sin N\beta}{\sin \beta} \right)^2 = \frac{N^2}{(N^2 - 1) \sin^2 \beta + 1}$$

Put above value in (1)

$$I_s = \left(\frac{A \sin \alpha}{\alpha} \right)^2 \times \frac{N^2}{(N^2 - 1) \sin^2 \beta + 1} \quad \text{--- (3)}$$

Divide (3) by (2)

$$\frac{I_s}{I_{\max}} = \frac{1}{(N^2 - 1) \sin^2 \beta + 1}$$

First principal Maxima

Secondary Maxima

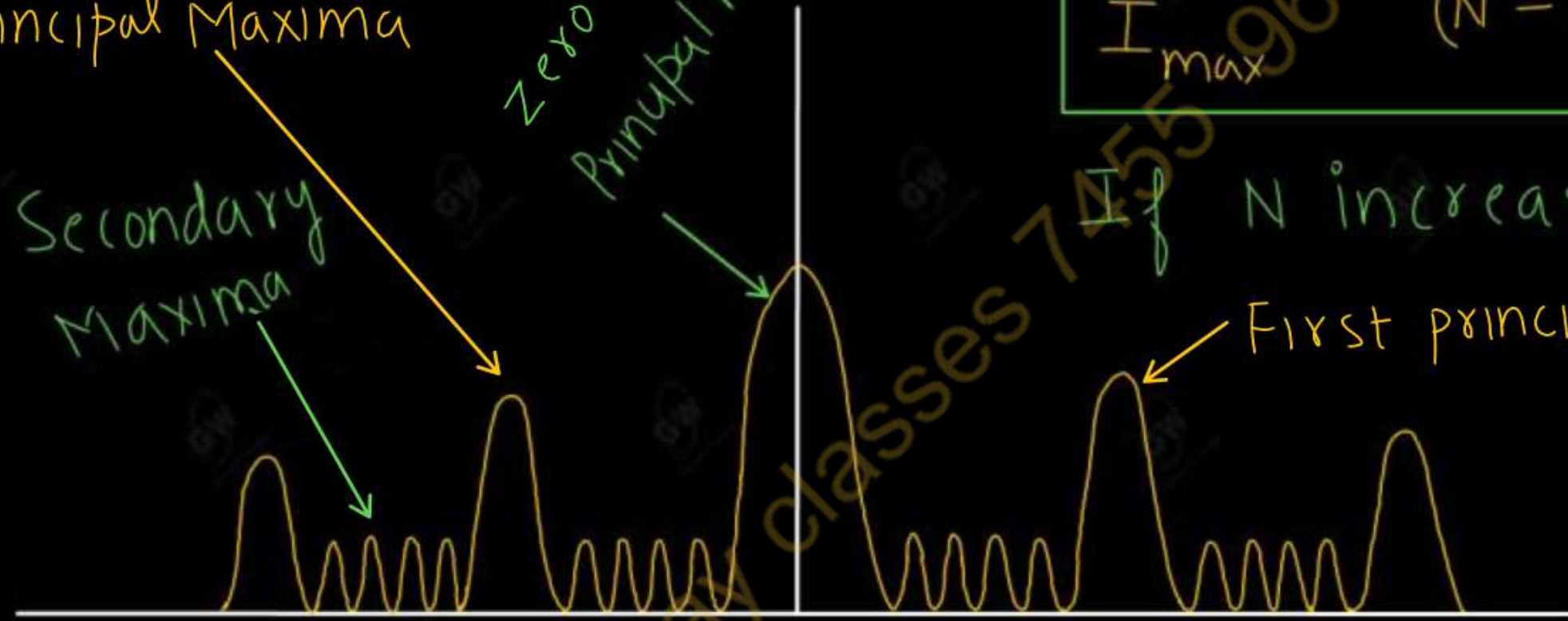
Zero order Principal maxima

If N increases, then I_s decreases

First principal maxima

Note

Total Number of Secondary maxima
 $(N - 2)$



$$I = \frac{A^2 \sin^2 \alpha}{\alpha^2} \times \frac{\sin^2 N\beta}{\sin^2 \beta}$$

Intensity due to diffraction
at a single slit

Intensity due to interference of light
waves from all N slits

- If in a particular direction $\frac{\sin^2 N\beta}{\sin^2 \beta}$ is maximum and $\frac{\sin^2 \alpha}{\alpha^2}$ is zero, then principle maxima will be absent from that direction.

- Direction of principle maxima

$$(e + d) \sin \theta = \pm n\lambda$$

$$n = 0, 1, 2, 3 \dots \dots \dots$$

- Direction of minima due to single slit

$$e \sin \theta = \pm m\lambda$$

$$m = 1, 2, 3 \dots \dots \dots N-1$$

- Divide (1) by (2)

$$\frac{(e+d) \sin \theta}{e \sin \theta} = \frac{n\lambda}{m\lambda}$$

$$\frac{e+d}{e} = \frac{n}{m}$$

This is the required condition for missing order spectra in diffraction grating.

(i) If $e = d$

$$\frac{e+e}{e} = \frac{n}{m}$$

$$n = 2m$$

$$m = 1, 2, 3 \dots$$

$$n = 2, 4, 6 \dots \dots \dots (\text{Missing order})$$

This means that 2, 4, 6, Orders of the interference maxima will be missing in the diffraction pattern.

(ii) If $2e = d$

$$\frac{e+2e}{e} = \frac{n}{m}$$

$$n = 3m$$

$$m = 1, 2, 3 \dots$$

$$n = 3, 6, 9 \dots \dots \dots (\text{Missing order})$$

This means that 3, 6, 9, Orders of the interference maxima will be missing in the diffraction pattern.

Formation of spectra by diffraction grating

The direction of n^{th} principle maxima is given by

$$(e + d) \sin \theta = \pm n \lambda$$

Where

$e + d = \text{Grating element}$

$n = \text{order}$

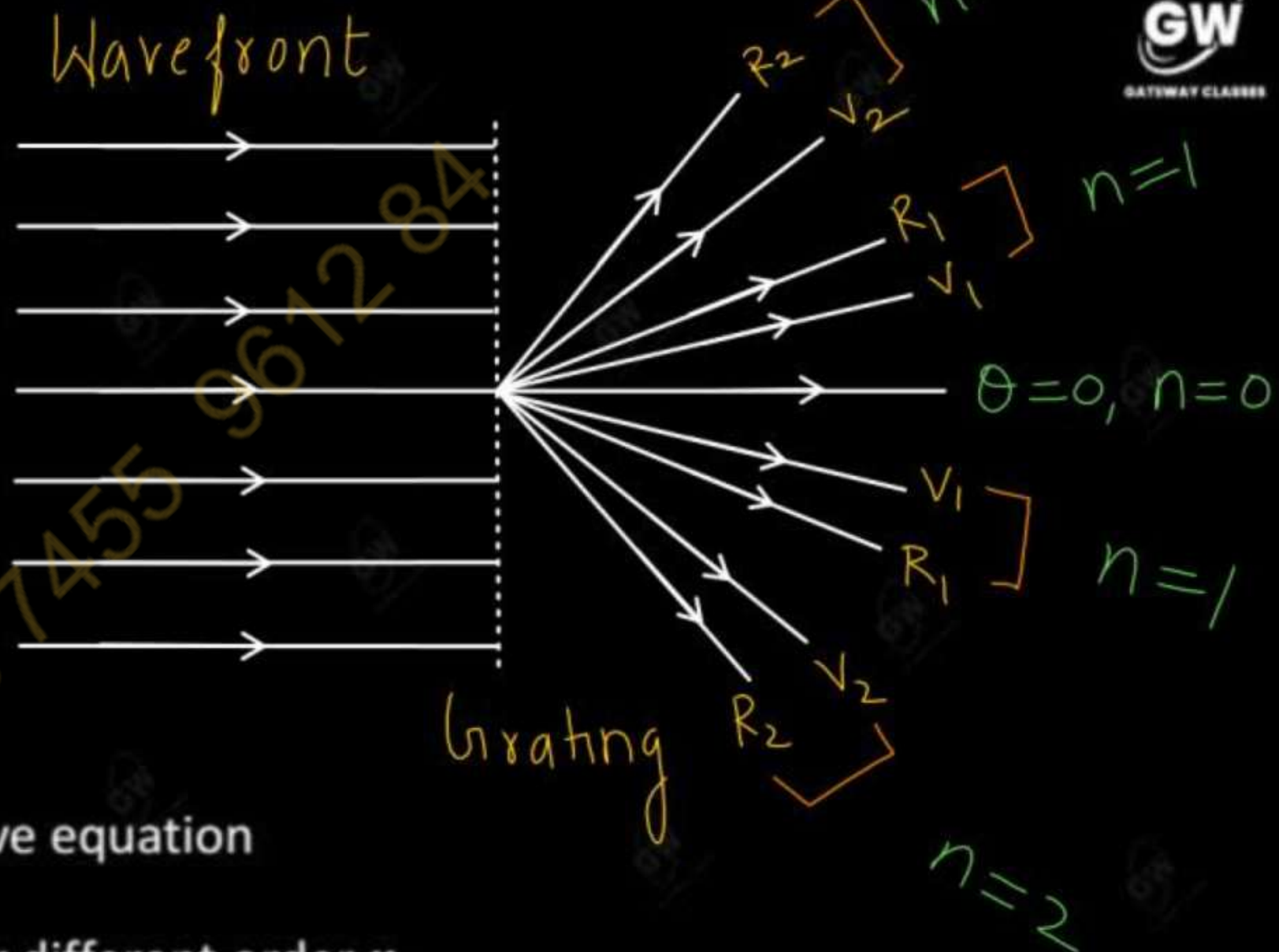
$\lambda = \text{Wavelength of light}$

$\theta = \text{Angle of diffraction}$

The following conclusion may be drawn from the above equation

- (i) For fixed value of $(e + d)$ and λ , θ is different for different order n .
- (ii) For fixed value of $(e + d)$ and n , θ is different for different wavelength

As $\lambda_r > \lambda_v$ then $\theta_r > \theta_v$



(iii) For $n = 0$, $\theta = 0$, zero order principal maxima for all values of λ lie in the same direction, Hence zero order principal maxima will be white.

Thus, the spectrum consist of white principal maxima of zero order having on either side of it the first order spectra, the second order spectra and so on. The spectra of each order consist of spectral colours in the order from violet to red.

Gateway classes 7455 9672 84

Fraunhofer diffraction due to N slits

Q.1 A parallel beam of monochromatic light is allowed to be incident normally on a plane grating having 1250 lines per cm and a second order spectral line is observed to be deviated through 30° . Calculate the wavelength of the spectral line.

Given

$$e+d = \frac{1}{1250} \text{ cm}$$

$$n = 2$$

$$\theta = 30^\circ$$

Using grating equation

$$(e+d) \sin \theta = n \lambda$$

$$\lambda = \frac{(e+d) \sin \theta}{n}$$

$$\lambda = \frac{1}{1250} \times \frac{\sin 30}{2}$$

$$\lambda = \frac{1}{1250} \times \frac{1}{2 \times 2}$$

$$\lambda = \frac{1}{5000}$$

$$\lambda = 2 \times 10^{-4}$$

$$\lambda = 2 \times 10^{-4} \times 10^8 \text{ \AA}$$

$$\lambda = 20000 \text{ \AA}$$

Q.2 In a plane transmission grating the angle of diffraction for the second order principal maxima for $\lambda = 5 \times 10^{-5} \text{ cm}$ is 30° . Calculate the number of lines in one cm of the grating surface.

Given

$$n = 2$$

$$\lambda = 5 \times 10^{-5} \text{ cm}$$

$$\theta = 30^\circ$$

Using grating equation

$$(e+d)\sin\theta = n\lambda$$

$$e+d = \frac{n\lambda}{\sin\theta}$$

$$= \frac{2 \times 5 \times 10^{-5}}{\sin 30}$$

$$e+d = 2 \times 10^{-4}$$

$$\frac{1}{e+d} = \frac{1}{2 \times 10^{-4}}$$

$$\frac{1}{e+d} = 5000$$

$$\begin{aligned} \text{Number of lines in 1 cm} \\ = 5000 \end{aligned}$$

Q.4 Monochromatic light from a Helium – neon Laser ($\lambda = 6328 \text{ \AA}$) is incident normally on a diffraction grating 6000 lines cm. Find the angles at which one would observe the first order maximum, the second order maximum

Given

$$\lambda = 6328 \text{ \AA}$$

$$e+d = \frac{1}{6000} \text{ cm}$$

Using grating equation

$$(e+d)\sin\theta = n\lambda$$

$$\sin\theta = \frac{n\lambda}{e+d}$$

For first order Maxima
 $n=1$

$$\sin\theta = \frac{1 \times 6328 \times 10^{-8}}{\frac{1}{6000}}$$

$$\sin\theta = 600 \times 6328 \times 10^{-8}$$

$$\theta = 22.31^\circ$$

For Second order Maxima
 $n=2$

$$\sin\theta = \frac{2 \times 6328 \times 10^{-8}}{\frac{1}{6000}}$$

$$\sin\theta = 2 \times 6328 \times 10^{-8} \times 6000$$

$$\theta = 49.41^\circ$$

Q.5 Parallel light consisting of two monochromatic radiations of $\lambda_1 = 6 \times 10^{-5} \text{ cm}$ and $\lambda_2 = 4 \times 10^{-5} \text{ cm}$ falls normally on a plane transmission grating ruled with 5000 lines per cm. What is the angular separation of the second order spectra of the two wavelengths?

Given

$$\lambda_1 = 6 \times 10^{-5} \text{ cm}$$

$$\lambda_2 = 4 \times 10^{-5} \text{ cm}$$

$$e+d = \frac{1}{5000} \text{ cm}$$

$$n = 2$$

For First Radiation

$$(e+d) \sin \theta_1 = n \lambda_1$$

$$\sin \theta_1 = \frac{n \lambda_1}{e+d}$$

$$\sin \theta_1 = \frac{2 \times 6 \times 10^{-5}}{\frac{1}{5000}}$$

$$\sin \theta_1 = 2 \times 6 \times 10^{-5} \times 5000$$

$$\theta_1 = 36.87^\circ$$

For Second Radiation

$$(e+d) \sin \theta_2 = n \lambda_2$$

$$\sin \theta_2 = \frac{n \lambda_2}{e+d} = \frac{2 \times 4 \times 10^{-5}}{\frac{1}{5000}}$$

$$\sin \theta_2 = 2 \times 4 \times 10^{-5} \times 5000$$

$$\theta_2 = 23.58^\circ$$

$$\begin{aligned} \text{Angular separation} &= \theta_1 - \theta_2 \\ &= 13.29^\circ \end{aligned}$$

Q.6 A diffraction grating used at normal incidence gives a yellow line ($\lambda = 6000 \text{ \AA}$) in a certain spectral order superimposed on a blue line ($\lambda = 4800 \text{ \AA}$) of next higher order.

If the angle of diffraction is $\sin^{-1}(3/4)$, calculate the grating element.

Given

$$\lambda_1 = 6000 \text{ \AA}$$

$$\lambda_2 = 4800 \text{ \AA}$$

$$\theta = \sin^{-1}\left(\frac{3}{4}\right)$$

$$\Rightarrow \sin \theta = \frac{3}{4}$$

If n^{th} order of λ_1 coincide with $(n+1)^{\text{th}}$ order of λ_2 then

$$(e+d) \sin \theta = n \lambda_1 \quad \text{--- (1)}$$

$$(e+d) \sin \theta = (n+1) \lambda_2 \quad \text{--- (2)}$$

From (1) and (2)

$$n \lambda_1 = (n+1) \lambda_2$$

$$n \lambda_1 = n \lambda_2 + \lambda_2$$

$$n \lambda_1 - n \lambda_2 = \lambda_2$$

$$n = \frac{\lambda_2}{\lambda_1 - \lambda_2} = \frac{4800}{6000 - 4800}$$

$$\boxed{n = 4}$$

Put n in (1)

$$e+d = \frac{n \lambda_1}{\sin \theta} = \frac{4 \times 6000 \times 10^{-10}}{3/4}$$

$$\boxed{e+d = 3.2 \times 10^{-6} \text{ m}}$$

Q.7 A diffraction grating used at normal incidence gives a green line ($5400 \text{ \AA}$) in a certain order n superimposed on the violet line ($4050 \text{ \AA}$) of the next higher order. If the angle of diffraction is 30° . Calculate the value of n . Also find how many lines per cm are there in the grating?

Given

$$\lambda_1 = 5400 \text{ \AA}$$

$$\lambda_2 = 4050 \text{ \AA}$$

$$\theta = 30^\circ$$

If n^{th} order of λ_1 coincide with $(n+1)^{\text{th}}$ order of λ_2 then

$$(e+d) \sin \theta = n \lambda_1 \quad \text{--- ①}$$

$$(e+d) \sin \theta = (n+1) \lambda_2$$

From ① and ②

$$n \lambda_1 = (n+1) \lambda_2$$

$$n \lambda_1 = n \lambda_2 + \lambda_2$$

$$n \lambda_1 - n \lambda_2 = \lambda_2$$

$$n = \frac{\lambda_2}{\lambda_1 - \lambda_2}$$

$$n = \frac{4050}{5400 - 4050}$$

$$n = \frac{4050}{1350}^3$$

$$n = 3$$

Put $n = 3$ in ①

$$(e+d)\sin\theta = n\lambda_1$$

$$e+d = \frac{n\lambda_1}{\sin\theta}$$

$$e+d = \frac{3 \times 5400 \times 10^{-8}}{\sin 30}$$

$$e+d = 6 \times 5400 \times 10^{-8}$$

$$\frac{1}{e+d} = \frac{1}{6 \times 5400 \times 10^{-8}}$$

$$= 3086$$

Number of lines per cm = 3086

Q.8 In a grating spectrum, which spectral line in 4th order will overlap with 3rd order line of 5461 Å

Given

$$n = 3$$

$$n+1 = 4$$

$$\lambda_1 = 5461 \text{ Å}$$

If n^{th} order of λ_1 coincide with $(n+1)^{\text{th}}$ order of λ_2 , then by grating equation

$$(e+d)\sin\theta = n\lambda_1 \quad \text{--- (1)}$$

$$(e+d)\sin\theta = (n+1)\lambda_2 \quad \text{--- (2)}$$

From (1) and (2)

$$n\lambda_1 = (n+1)\lambda_2$$

$$3 \times \lambda_1 = 4 \lambda_2$$

$$\lambda_2 = \frac{3}{4} \lambda_1$$

$$\lambda_2 = \frac{3}{4} \times 5461$$

$$\lambda_2 = 4095.75 \text{ Å}$$

Diffraction

Q.1 What do you understand by Grating?

Q.2 Give the construction and theory of plane diffraction grating

OR

Give the construction and theory of plane transmission grating

OR

Discuss the phenomenon of diffraction due to plane diffraction grating

Q.3 Explain the formation of spectra by diffraction grating.

Q.4 What do you understand by missing or absent order spectrum? What particular spectra would be absent if the width of transparencies twice of opacities of grating

Q.5 What particular spectra would be absent if the width of transparencies and opacities of the grating are equal



B. Tech : Engg. Physics



Unit : Wave Optics

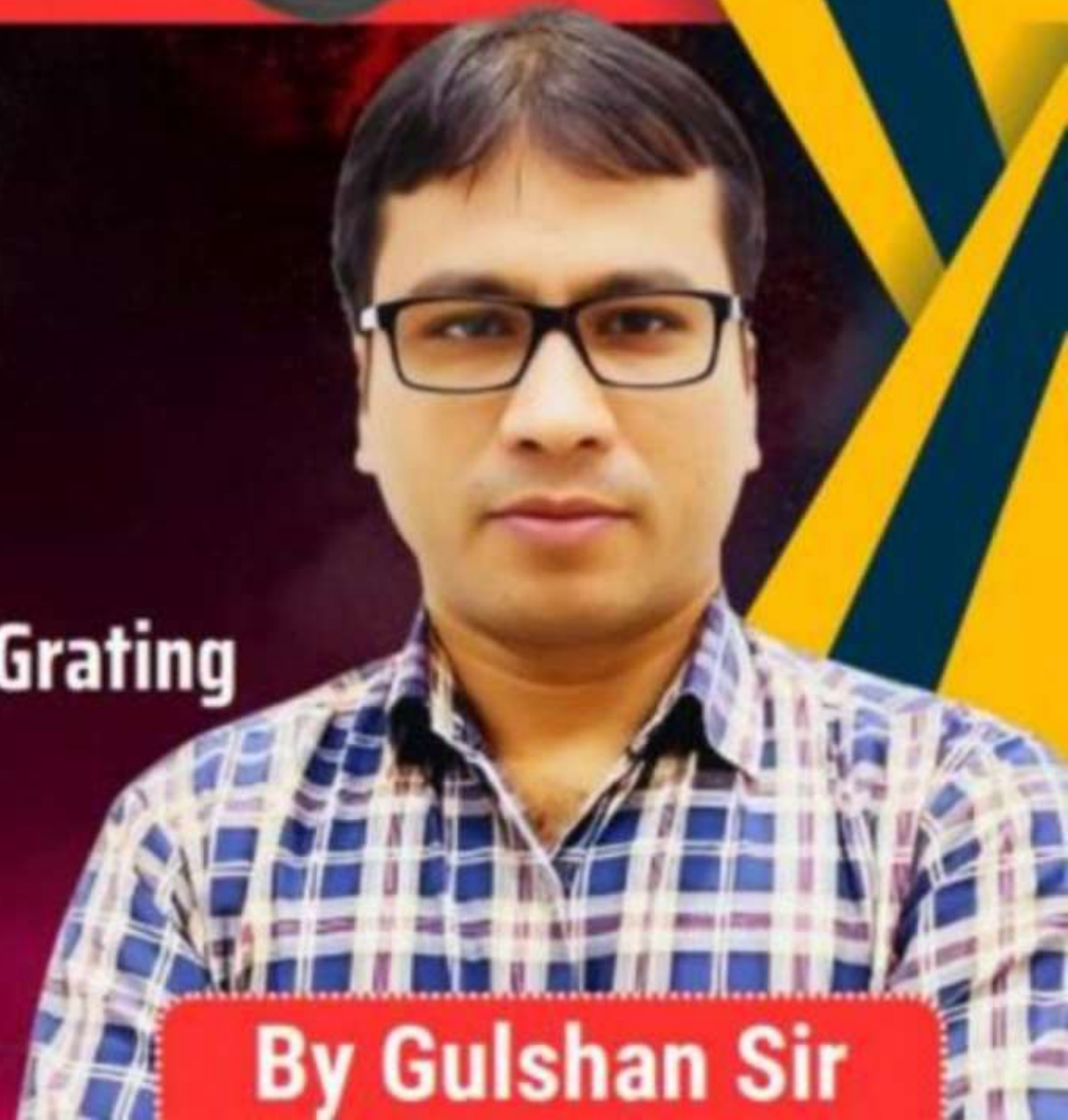


(Diffraction)

Lecture - 8

Today's Target

- ✓ Dispersive Power, Resolving Power
- ✓ Rayleigh's criteria of Resolution, Resolving Power of Grating
- ✓ DPP
- ✓ PYQs



By Gulshan Sir

The dispersive power of a diffraction grating is defined as the rate of change of the angle of diffraction with the change in the wavelength of light

$$\text{Dispersive power} = \frac{d\theta}{d\lambda}$$

Grating equation for plane diffraction grating

$$(e+d)\sin\theta = n\lambda \quad \text{--- ①}$$

diff. wrt λ

$$(e+d)\cos\theta \frac{d\theta}{d\lambda} = n$$

$$\frac{d\theta}{d\lambda} = \frac{n}{(e+d)\cos\theta}$$

$$\frac{d\theta}{d\lambda} = \frac{n}{(e+d)\sqrt{1-\sin^2\theta}}$$

$$\text{But } \sin\theta = \frac{n\lambda}{e+d}$$

$$\frac{d\theta}{d\lambda} = \frac{n}{(e+d)\sqrt{1-\left(\frac{n\lambda}{e+d}\right)^2}}$$

$$\frac{d\theta}{d\lambda} = \frac{n}{\sqrt{(e+d)^2 - (n\lambda)^2}}$$

$$\frac{d\theta}{d\lambda} = \frac{n}{\sqrt{\left(\frac{e+d}{n}\right)^2 - \lambda^2}}$$

$$\frac{d\theta}{d\lambda} = \frac{1}{\sqrt{\left(\frac{e+d}{n}\right)^2 - \lambda^2}}$$

✓

Conclusion

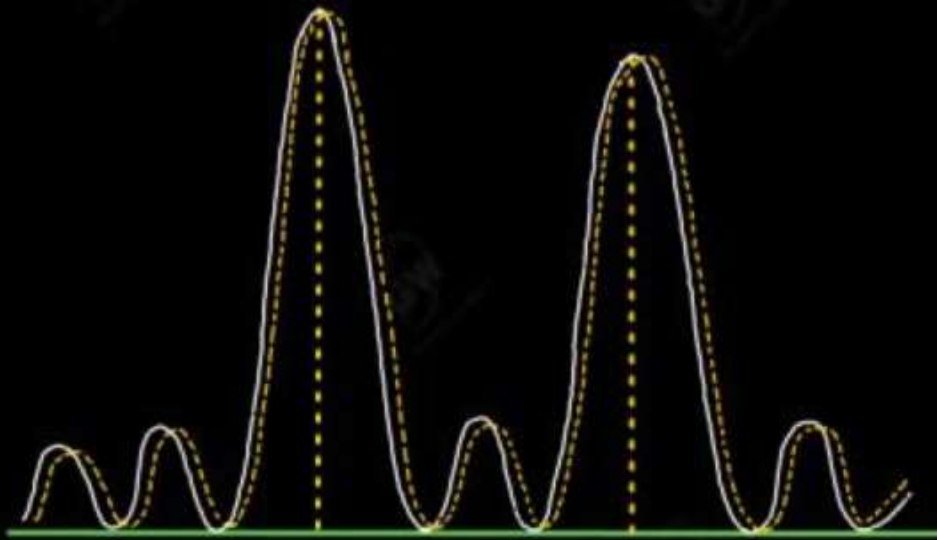
- The Dispersive power is directly proportional to the order of spectrum.
- The dispersive power is inversely proportional to the grating element ($e + d$).
- The dispersive power is inversely proportional to $\cos\theta$ (i.e. larger the value of θ higher is the dispersive power)

Resolving power of an optical instrument

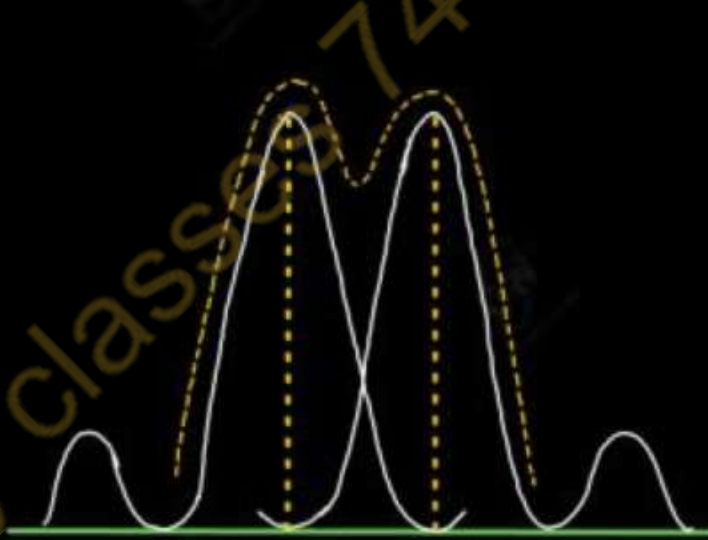
- The capacity of an optical instrument to show two close objects separately is called resolution and the ability of an optical instrument to just resolve the images of two close point objects is called its Resolving power.

The Rayleigh criterion of Resolution

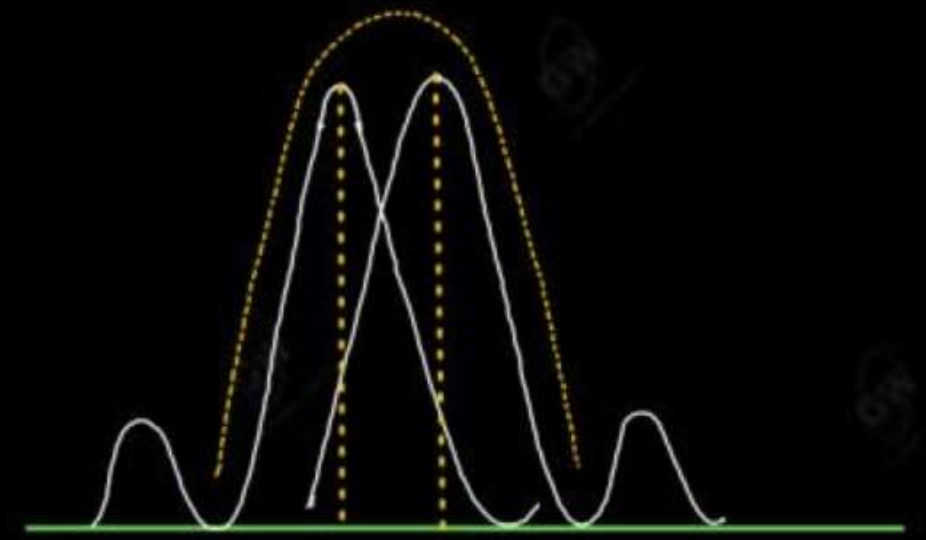
According to Rayleigh's criterion, the two close point objects or spectral lines are just resolved (i.e. seen separately) by an optical instrument, when the principal maximum of the diffraction pattern of one object coincide with the first minimum of the diffraction pattern of the other object and vice-versa.



Fully Resolved



Just Resolved



Not Resolved

- This criterion is applicable for the evaluation of resolving power of optical instrument like Telescope, Microscope etc.

Resolving Power of Grating

It is defined as the ability of a grating to form two well separated diffraction patterns corresponding to two close wavelengths. If a grating can just resolve the spectral lines of wavelength λ and $\lambda + d\lambda$, then its resolving power is expressed as

$$\text{Resolving power of grating} = \frac{\lambda}{d\lambda}$$

- Let AB represent plane transmission grating having grating element $(e + d)$ and total number of slits N .



- Consider a beam of light having two wavelength λ and $\lambda + d\lambda$ incident on grating

The direction of n^{th} principal maxima for a spectral line of wavelength (λ)

$$(e+d)\sin\theta = n\lambda \quad \text{--- (1)}$$

$$n = 0, 1, 2, 3, \dots$$

First minima adjacent to n^{th} principal maxima will be formed in the direction

$\theta + d\theta$ as

$$N(e+d)\sin(\theta + d\theta) = m\lambda \quad \text{--- (2)}$$

Where

$$m = 1, 2, 3, \dots, N-1$$

$$m \neq 0, N, 2N, 3N, \dots$$

Hence

$$m = nN + 1$$

Put m in (2)

$$N(e+d)\sin(\theta + d\theta) = (nN + 1)\lambda$$

$$\cancel{N}(e+d)\sin(\theta + d\theta) = \cancel{N}\left(n + \frac{1}{N}\right)\lambda$$

$$(e+d)\sin(\theta + d\theta) = \left(n + \frac{1}{N}\right)\lambda$$

$$\text{--- (3)}$$

The direction of n^{th} principal maxima for the Spectral line of wavelength $(\lambda + d\lambda)$

$$(e+d)\sin\theta = n(\lambda + d\lambda)$$

According to Rayleigh's criterion of resolution, n^{th} principal maxima of $(\lambda + d\lambda)$ coincide with the first minima of λ in the direction $\theta + d\theta$

$$(e+d)\sin(\theta + d\theta) = n(\lambda + d\lambda) \quad \text{--- (4)}$$

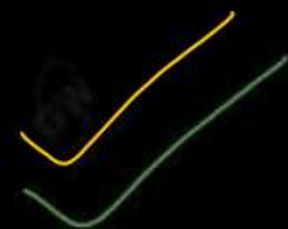
On comparing eqⁿ (3) and (4)

$$\left(n + \frac{1}{N}\right)\lambda = n(\lambda + d\lambda)$$

$$n\lambda + \frac{\lambda}{N} = n\lambda + nd\lambda$$

$$\frac{\lambda}{N} = nd\lambda$$

$$\boxed{\frac{\lambda}{d\lambda} = Nn}$$



Thus Resolving power is directly Proportional to

- (i) The order of Spectrum (n)
- (ii) Total number of lines on the grating

Relation between Dispersive power and Resolving power of Grating

Dispersive Power

$$\frac{d\theta}{d\lambda} = \frac{n}{(e+d)\cos\theta} \quad \text{--- ①}$$

Resolving power

$$\frac{\lambda}{d\lambda} = Nn$$

$$\frac{\lambda}{d\lambda} = N \times \frac{n}{(e+d)\cos\theta} \times (e+d)\cos\theta \quad \text{--- ②}$$

Using ① in ②

$$\frac{\lambda}{d\lambda} = N \times \frac{d\theta}{d\lambda} \times (e+d)\cos\theta$$

$$\frac{\lambda}{d\lambda} = N(e+d)\cos\theta \frac{d\theta}{d\lambda}$$

Where

$N(e+d)\cos\theta \rightarrow$ Total Aperture

Resolving power = Total Aperture $\times$ Dispersive power

Q.1 What must be minimum number of lines per cm in a half inch width grating to resolve the wavelength 5890 Å and 5896 Å?

Given

$$\lambda_1 = 5890 \text{ Å}$$

$$\lambda_2 = 5896 \text{ Å}$$

$$n = 1$$

$$d\lambda = \lambda_2 - \lambda_1$$

$$d\lambda = 5896 - 5890$$

$$d\lambda = 6 \text{ Å}$$

$$\lambda = \frac{\lambda_1 + \lambda_2}{2}$$

$$\lambda = \frac{5890 + 5896}{2}$$

$$\lambda = 5893 \text{ Å}$$

Resolving power

$$\frac{\lambda}{d\lambda} = nN$$

Minimum Number of lines in Half inch

$$N = \frac{1}{n} \left(\frac{\lambda}{d\lambda} \right)$$

$$= \frac{1}{1} \left(\frac{5893}{6} \right) = 982$$

Minimum Number of lines per inch

$$N = 982 \times 2$$

$$N = 1964$$

Minimum Number of lines per cm

$$N = \frac{1964}{2.54}$$

$$N = 773$$

Q.2 A plane transmission grating has 15000 lines per inch. Find the resolving power of the grating and the smallest wavelength difference that can be resolved with a light of wavelength $6000 \text{ \AA}$ in second order

Given

$$N = 15000$$

$$\lambda = 6000 \text{ \AA}$$

$$n = 2$$

Resolving Power

$$\frac{\lambda}{d\lambda} = nN$$

$$= 2 \times 15000$$

$$= 30,000$$

Smallest wavelength difference

$$d\lambda = \frac{\lambda}{nN}$$

$$d\lambda = \frac{6000}{30,000}$$

$$d\lambda = 0.2 \text{ \AA}$$

Q.3 A plane transmission grating has 16000 lines to an inch over a length of 5 inches.

Find in the wavelength region of $6000 \text{ \AA}$, in the second order

- (i) the resolving power of the grating and
- (ii) the smallest wavelength difference that can be resolved.

Given

$$N = 16000 \times 5$$

$$= 80,000$$

$$\lambda = 6000 \text{ \AA}$$

$$n = 2$$

Resolving Power

$$\frac{\lambda}{d\lambda} = Nn$$

$$= 80,000 \times 2$$

$$= 160,000$$

Smallest Wavelength Difference

$$d\lambda = \frac{\lambda}{Nn}$$

$$= \frac{6000}{160,000}$$

$$d\lambda = 0.0375 \text{ \AA}$$

Q.4 Find the angular separation of 5048 Å and 5016 Å wavelength in second order spectrum

obtained by a plane diffraction grating having 15000 line per inch.

Given

$$\lambda_1 = 5048 \text{ Å}$$

$$\lambda_2 = 5016 \text{ Å}$$

$$n = 2$$

$$N = 15000$$

$$e+d = \frac{2.54}{15000} \text{ cm}$$

Dispersive power

$$\frac{d\theta}{d\lambda} = \frac{n}{\sqrt{(e+d)^2 - (n\lambda)^2}}$$

$$d\theta = \frac{nd\lambda}{\sqrt{(e+d)^2 - (n\lambda)^2}}$$

$$d\theta \approx \frac{nd\lambda}{e+d}$$

$$d\theta = \frac{2 \times (5048 - 5016) \times 10^{-8}}{\frac{2.54}{15000}}$$

$$= \frac{2 \times 32 \times 15000 \times 10^{-8}}{2.54}$$

$$d\theta = 3.79 \times 10^{-3} \text{ radian}$$

UNIT : Wave Optics

✓ Q.1 What is dispersive power of plane transmission grating ?

✓ Q.2 What do you mean by Resolving power of an optical instrument ?

✓ Q.3 What is Resolving power of grating ?

✓ Q.4 What is the Rayleigh's criterion of resolution how one can increase the resolving power of a diffraction grating? Using Rayleigh's criterion for just resolution show that the resolving power of grating is equal to nN , where n is the order of spectrum and N is the total number of lines on the grating.

✓ Q.5 Explain briefly the Rayleigh's criterion of resolution. Discuss the resolving power of plain transmission grating and find the relation between resolving and dispersive power of the grating

AKTU Full Courses (Paid)
Download **Gateway Classes** Application
From Google Play store
All Subjects

Link in Description

Thank You

AKTU



B. Tech I-Year

Gateway Classes



Full Courses Available in App

AKTU B.Tech I- Year : All Branches

AKTU B.Tech II- Year

Branches : 1. CS IT & Allied

2. EC & Allied

3. ME & Allied

4. EE & Allied

Downlod App Now



**Full
Courses**

✓ **V. Lectures**

✓ **Pdf Notes**

✓ **AKTU PYQs**

✓ **DPP**